

Simultaneous Physical Constraint Collapse of the Neutrino Carrier Universe

Shared-witness survival across propagation, gauge, anomaly, vacuum, effective-theory, and generalized-symmetry interfaces

Principal result

Within the fixed neutrino-oscillation target and comparison class, simultaneous consistency defines a proper, nonempty physical residue of the complete source-first carrier universe:

$$\text{CarrierSurv}_{\text{sem}}(T_{\nu}^{(4)})/\equiv_{\text{car}} \cong \mathfrak{S}_{\nu}^{(4)}, \quad \emptyset \neq \mathfrak{S}_{\nu}^{(4)} \subsetneq \mathfrak{U}_{\nu}.$$

The undeformed massless-neutrino Standard Model and the pure topological, pure defect, pure boundary, and pure environmental strata are excluded as stand-alone realizations. Gauge-invariant law-only and local/hidden completions furnish carrier-distinct surviving classes. The result establishes simultaneous survivor closure, not unique occupancy or physical realization.

Amos Jay Maley
Independent Researcher

15 August 2026

Abstract

Previous target-relative results isolate an ordered neutrino-propagation role complex, determine its complete carrier-neutral physical interface, and generate a source-first carrier universe independently of interface success. The present paper addresses the remaining simultaneous-consistency problem: which carrier architectures realize the full interface while satisfying every active propagation, gauge, anomaly, vacuum, effective-theory, generalized-symmetry, and co-instantiation condition in one physical completion?

The analysis first derives a complete survival-facing trace and an independent many-sorted survival-consistency theory. Intrinsic conditions, cross-signature relations, and one-completion/common-cone requirements are then shown to admit a complete constraint normal form. Each source family is evaluated by an explicit map containing the event-complete instrument, charged-current attachment, active-transfer kernel, visible response classes, causal and environmental support, representation and charge data, anomaly residuals, vacuum and deformation moduli, effective-theory matching, activated extended-sector information, and the complete physical overlap structure. Exact survivor loci are therefore preimages of a global semantic model locus rather than products of marginal pass conditions.

The resulting collapse is nontrivial. The undeformed massless-neutrino Standard Model stratum is empty. Pure topological, defect, boundary, and environmental carriers cannot realize the full source-to-detection response on their own, although they remain admissible as active co-carriers in dynamical hybrids. A gauge-invariant dimension-five law-only completion and gauge-singlet local/hidden completions give explicit carrier-distinct survivor witnesses. Composite, solitonic, hidden, and hybrid families are retained through exact symbolic moduli loci. Consequently,

$$\emptyset \neq \mathfrak{S}_\nu^{(4)} \subsetneq \mathfrak{U}_\nu, \quad \text{CarrierSurv}_{\text{sem}}(T_\nu^{(4)})/\equiv_{\text{car}} \cong \mathfrak{S}_\nu^{(4)}.$$

The surviving residue is not singleton at the carrier-semantic level. The theorem therefore does not select a unique mechanism or imply physical realization; compatible role occupancy remains a distinct subsequent question.

Keywords. admissibility; neutrino oscillation; simultaneous physical realization; carrier universe; shared-witness pullback; anomaly cancellation; vacuum compatibility; effective field theory; generalized symmetry; survivor completeness.

Scope and interpretation

The result is an exact theorem relative to a fixed target, source ontology, comparison class, and previously established role/interface/carrier results. It is not a ranking of named neutrino models. A symbolic family locus counts as evaluated only when it is specified by the physical equations, cross-coordinate relations, positivity or existence conditions, and common-witness constraints defining the global semantic model locus. The supplementary tables supplied with the source package are navigational representations of these results; the scientific claims are established by the definitions and proofs in the manuscript. Standard channel, anomaly, effective-theory, and representative completion results are cited from the primary literature, while the target-relative structural results are imported from the preceding papers in this series.

Contents

Abstract

Scope and interpretation

1	Introduction	1
1.1	From role necessity to simultaneous survival	1
1.2	Why componentwise consistency is insufficient	1
1.3	Principal constructions	1
1.4	Main results	1
1.5	Interpretive boundary	2
1.6	Organization	2
2	Fixed inputs, target, and scope invariance	2
2.1	Inherited role structure	2
2.2	Inherited physical interface	3
2.3	Inherited carrier universe	3
2.4	Target object	3
2.5	Comparison class	3
2.6	Scope changes and premise revision	3
3	Dependency structure, source roles, and primary-physics boundary	4
3.1	Dependency stack	4
3.2	Source-role matrix	4
3.3	Primary-source physics boundary	5
3.4	Proof hierarchy	5
3.5	Load-bearing dependency order	5
3.6	Primary-source versus corpus-source use	6
3.7	No evidential inheritance by repetition	6
4	Semantic survivor class	6
4.1	Why the semantic prior is required	6
4.2	Survival trace	7
4.3	Semantic survival as one-object physical realization	7
4.4	Role completeness inside the semantic definition	8
4.5	Target-relative rather than world-relative identity	8
4.6	Falsifiability of the semantic prior	8
5	Survival-consistency theory and evaluator completeness	8
5.1	Why relations must be fixed before the constraint list	8
5.2	Raw survival presentations and the survival-facing trace	8
5.3	The many-sorted survival signature	9
5.4	Semantic relation class before finite normal form	9
5.5	Semantic uses and survival evaluators	11
5.6	Finite constraint normal form	11

6	Candidate evaluation	13
6.1	Disposition type	13
6.2	Totality	13
6.3	Evaluation congruence	13
6.4	Evaluation independence	13
6.5	Evaluation as a typed partial-to-total completion	14
6.6	Exactness over symbolic parameter domains	14
6.7	Carrier-equivalence invariance	14
6.8	Diagnostic quantities and standing-relevant output	15
7	Common physical witness structure	15
7.1	Coordinate meanings	15
7.2	Overlap structure	16
7.3	Physical identity in the witness diagram	16
7.4	Minimal overlap requirements	17
7.5	No metadata promotion	17
7.6	Witness packet as PCR-5 evidence	17
8	Simultaneous constraint normal form	17
8.1	Constraint family	18
8.2	Constraint completeness	18
8.3	Failure decomposition	19
8.4	Why the bundle is complete but not universal	19
8.5	Constraint interaction	19
8.6	No optimization principle	20
9	Raw, co-instantiated, and survivor moduli	20
9.1	Raw evaluation object	20
9.2	Pullback and survivor objects	20
9.3	Relational interpretation of the pullback	21
9.4	Failure of a hidden Helly principle	21
9.5	Relation to fiber products over the target trace	21
9.6	Survivor object as a scientific residue	21
10	Nested role, charged-current boundary, channel, and response constraints	21
10.1	Role capability as one-architecture burden	21
10.2	Charged-current boundary	22
10.3	Event-complete channel	22
10.4	Visible response multiplicity	22
10.5	Channel-native formulation across microscopic realizations	22
10.6	Environmental dependence of the response packet	23
10.7	No mass-language shortcut	23
11	Locality, transport, causality, and environmental participation	23
11.1	Transport object	23
11.2	Environmental and medium co-carriers	23
11.3	Causal discipline	24
11.4	Support categories and target locality	24
11.5	Environment as carrier rather than calibration	24
11.6	No environment patchwork	24

12 Gauge attachment, representation, charge, and anomaly closure	24
12.1 Gauge attachment	25
12.2 Anomaly bundle	25
12.3 Global-form dependence	25
12.4 Gauge consistency versus global anomaly matching	25
12.5 Charge normalization and lawful equivalence	26
12.6 Gauge-vacuum-anomaly coupling	26
12.7 Global-form activation	26
13 Vacuum support, deformation strata, and EFT matching	26
13.1 Vacuum-support object	26
13.2 Parameter-stratified deformation	26
13.3 EFT matching	27
13.4 Decoupling	27
13.5 Vacuum existence versus vacuum compatibility	27
13.6 Deformation strata without naturalness	27
13.7 Matching preserves carrier responsibility	27
14 Global form, higher-form, higher-group, defect, and non-invertible relevance	28
14.1 Relevance gate	28
14.2 Non-invertible candidates	28
14.3 Four-way relevance disposition	28
14.4 Constitutive coherence versus target survival	29
14.5 Extended-sector changes after failure	29
15 Explicit survival maps and family specialization	29
15.1 Why symbolic does not mean unevaluated	29
15.2 Universal candidate data	29
15.3 Instrument and record residuals	29
15.4 Boundary and gauge attachment	30
15.5 Transport, environment, and causal support	30
15.6 Representation, charge, and anomaly residual	31
15.7 Vacuum and deformation moduli	31
15.8 EFT and matching moduli	31
15.9 Activated extension and simultaneous realization	32
15.10 Cross-signature relation residuals	32
15.11 Universal global evaluation map	33
16 Evaluated survivor loci I: existing-substrate, law-only, and local carriers	33
16.1 Family overlaps and the meaning of a pure stratum	33
16.2 Strict existing-substrate stratum	33
16.3 Law-only family	34
16.4 New local primitive carriers	35
16.5 Local anomaly equations are evaluated, not labelled	36
17 Evaluated survivor loci II: composite, topological, defect, solitonic, boundary, environmental, and hidden carriers	36
17.1 Composite and emergent carriers	36
17.2 Pure topological carriers	37
17.3 Pure extended and defect-supported carriers	37

17.4 Solitonic and localized nonperturbative carriers	37
17.5 Pure boundary and edge carriers	38
17.6 Pure environmental carriers	38
17.7 Hidden and enlarged-state-space carriers	39
18 Hybrid closure, actual survivor residue, and parameter-stratified survival	39
18.1 Hybrids as first-class candidates	40
18.2 Pure-family reductions	40
18.3 Actual evaluated survivor residue	40
18.4 Exact parameter strata	41
18.5 PCR-3 activity strata remain fixed	41
18.6 Dense and isolated survivor loci	42
19 Failure classification, fixed-candidate exclusion, and omission tests	42
19.1 Failure classes	42
19.2 Fixed-candidate failure	43
19.3 Omitted-candidate test	43
19.4 Failure localization	43
19.5 Classification of responses to failure	44
20 Survivor equivalence, interface degeneracy, and live residue	44
20.1 Inherited carrier equivalence	44
20.2 Interface equality is coarser	44
20.3 Presentation quotient and live residue	44
20.4 Automorphism preparation	45
20.5 Three equivalence layers	45
20.6 Residue geometry	45
21 Semantic completeness and the exact survivor theorem	45
21.1 Coverage and soundness maps	45
21.2 Why the theorem is not definitional	46
21.3 Falsifiers	47
21.4 Independent completeness burdens	47
21.5 No hidden survivor multiplicity	47
22 Robustness tests and scope boundaries	48
22.1 Marginal consistency versus a global physical realization	49
22.2 Operational scope of the propagation channel	49
22.3 Scope of source completeness	49
22.4 Use of empirical information	49
22.5 Further diagnostic constructions	49
22.6 Direct falsification criteria	50
23 Conclusion and consequences for role-occupancy analysis	50
23.1 Physical interpretation of the residue	50
23.2 Consequences for compatible role occupancy	51
23.3 Scope variation	51
23.4 Independence from later assignment structure	51
23.5 Final statement	51

A	Constraint normal forms and failure conditions	52
A.1	Inapplicability	53
A.2	Target changes	53
A.3	Independence from model preference	53
B	Family-level survivor loci	53
B.1	Universal signature matrix	54
B.2	Actual residue	54
C	Provenance of imported results	55
D	Supplementary proofs for simultaneous constraint closure	55
D.1	Evaluation totality as a sum-type theorem	55
D.2	Constraint completeness from the independent relation theory	56
D.3	Why common-cone existence is not relation completeness	56
D.4	Specialization to family parameter spaces	56
D.5	Positive-witness relation closure	56
D.6	Common-cone invariance	56
D.7	Pairwise insufficiency in diagram form	56
D.8	Fail-closedness and candidate identity	57
D.9	Parameter-locus exactness	57
D.10	Properness and nonemptiness	57
E	Family-specialized evaluator maps	57
E.1	Existing-substrate family	57
E.2	Law-only family	57
E.3	Local-carrier family	58
E.4	Composite and emergent family	58
E.5	Topological and defect families	58
E.6	Solitonic family	58
E.7	Boundary and environmental families	58
E.8	Hidden/enlarged-state-space family	58
E.9	Hybrid family	58
	References	59

1 Introduction

1.1 From role necessity to simultaneous survival

The analysis begins from three previously established objects. The first is the ordered neutrino-propagation role complex

$$\mathcal{R}_\nu^{\text{osc}} = (\{r_{\triangleright\triangleleft}, r_\Delta\}, r_\Delta \succ r_{\triangleright\triangleleft}), \quad r_\Delta \Rightarrow r_{\triangleright\triangleleft}, \quad r_{\triangleright\triangleleft} \not\Rightarrow r_\Delta.$$

The second is the complete target-relative physical interface, including record, continuation, gauge, anomaly, vacuum, effective-theory, activated-extension, and physical co-instantiation data. The third is the source-first carrier universe

$$\text{CarrierReal}_{\text{sem}}(T_\nu^{(3)}) / \equiv_{\text{car}} \cong \mathfrak{U}_\nu,$$

whose source-coherent parameter points, activity reductions, and carrier identities are fixed independently of interface success [12, 16, 17].

The remaining problem is therefore neither candidate generation nor role assignment. It is the simultaneous physical intersection problem: for each independently generated carrier architecture, determine the exact parameter locus on which the complete interface and every active source, propagation, gauge, anomaly, vacuum, effective-theory, generalized-symmetry, and one-realization condition hold in one physical completion.

Problem addressed in this paper

For every carrier instance in $\mathfrak{U}_\nu$, determine the exact locus on which the full target interface and all active physical consistency conditions are realized by one common witness architecture, and identify the resulting survivor residue without introducing any new selection principle.

1.2 Why componentwise consistency is insufficient

A propagation channel, anomaly-cancellation pattern, vacuum state, and effective matching map can each be individually admissible while belonging to mutually incompatible physical constructions. Conversely, a finite list of marginal conditions cannot establish that every same-target cross-sector consistency relation has been included. The paper therefore separates two logically different tasks.

First, a complete survival-consistency theory is derived from the fixed target-facing trace, the source ontology, the known-substrate consistency relations, activated extension sectors, and one-completion semantics. Second, this theory is specialized to every source family by explicit evaluation maps, producing exact survivor loci rather than the circular set description “all points that pass.”

1.3 Principal constructions

For a carrier instance $x = [u, \theta] \in \mathfrak{U}_\nu$, let $\mathcal{W}_4(x)$ denote the complete candidate witness packet and let $\tau_4^{\text{surv}}(x)$ be its survival-facing trace. The independently defined survival-consistency theory is written

$$\mathbb{S}_4^{\text{surv}} = (\mathfrak{O}_4, \mathfrak{M}_4, \mathfrak{E}_4),$$

and its semantic model locus is $\mathcal{A}_4^{\text{sem}}$. For each source family F , the complete physical evaluation is encoded by

$$\widehat{\Phi}_F : \Theta_F^{\text{part}} \longrightarrow \mathcal{Y}_F,$$

with survivor locus

$$\Theta_F^{\text{surv}} = \widehat{\Phi}_F^{-1}(\mathcal{Z}_4^{\text{sem}}).$$

The evaluated residue is then the carrier-equivalence quotient of the union of these exact family loci.

1.4 Main results

Four results organize the paper.

First, the semantic survival-evaluator class is generated by the intrinsic, cross-signature, and one-completion relation schemas of the independent survival theory:

$$\text{PhysSurvEval}_4^{\text{sem}} = \text{Eval}(\mathbb{S}_4^{\text{surv}}).$$

Second, every source family admits an exact family-specific survivor locus

$$\Theta_F^{\text{surv}} = \widehat{\Phi}_F^{-1}(\mathcal{X}_4^{\text{sem}}).$$

Third, simultaneous consistency produces a proper and nonempty reduction of the complete carrier universe:

$$\emptyset \neq \mathfrak{S}_\nu^{(4)} \subseteq \mathfrak{U}_\nu.$$

The undeformed massless-neutrino Standard Model and the pure topological, pure defect, pure boundary, and pure environment strata are excluded as stand-alone realizations. A gauge-invariant law-only completion and active local/hidden singlet completions furnish explicit survivor witnesses.

Fourth, the semantic and canonical survivor residues coincide:

$$\boxed{\text{CarrierSurv}_{\text{sem}}(T_\nu^{(4)})/\equiv_{\text{car}} \cong \mathfrak{S}_\nu^{(4)} = \mathfrak{S}_{\nu, \text{eval}}^{(4)}}.$$

1.5 Interpretive boundary

Nonclaim 1.1 (Survival is not role occupancy). *The exact survivor residue does not imply that a survivor is a licensed occupant of $\mathcal{R}_\nu^{\text{osc}}$, still less that nature realizes it. PCR-5 must independently construct the role-eligibility relation, joint-assignment compatibility, live residue, and lawful role-incidence automorphism quotient required by Role-Occupancy Closure [15].*

The survivor theorem is therefore deliberately weaker than a mechanism-selection theorem and stronger than a collection of model-by-model viability statements. It identifies the exact carrier-semantic residue after simultaneous physical consistency, while leaving compatible role occupancy and realization to a distinct subsequent analysis.

1.6 Organization

Sections 2–5 establish the fixed setting, semantic survivor class, global survival-consistency theory, and evaluator completeness. Sections 6–9 define candidate evaluation, the common witness structure, the constraint normal form, and the pullback geometry of simultaneous realization. Sections 10–14 develop the physical interface conditions. Section 15 gives the universal and family-specialized evaluation maps. Sections 16–18 derive the exact family loci and the proper nonempty collapse. Sections 19–21 establish failure classification, quotient congruence, and the capstone correspondence. The final sections discuss robustness, scope boundaries, and the mathematical consequences for subsequent role-occupancy analysis.

2 Fixed inputs, target, and scope invariance

2.1 Inherited role structure

The present analysis takes as fixed the previously established ordered role complex and its target evidence:

$$\mathcal{R}_\nu^{\text{osc}} = (\{r_{\triangleright\triangleleft}, r_\Delta\}, r_\Delta \succ r_{\triangleright\triangleleft}). \quad (1)$$

It does not reinterpret r_Δ as mass, $r_{\triangleright\triangleleft}$ as a PMNS matrix, or either role as evidence for a particular carrier.

2.2 Inherited physical interface

The complete target-relative physical interface and its exact semantic quotient are taken from the preceding interface theorem. We use that result through the profile

$$\mathcal{P}_v^+ = (P_{\text{rec}}, C_{\text{cont}}, S_{\text{loc}}, G_{\text{attach}}, A_q, V_{\text{def}}, R_{\text{EFT}}, X_{\text{ext}}, \mathcal{J}_v^{\text{sim}}), \quad (2)$$

where $\mathcal{J}_v^{\text{sim}}$ records physical co-instantiation and common-cone structure. Equality of $\mathcal{P}_v^+$ is equivalent to target-semantic interface indistinguishability within one fixed target fiber [12].

2.3 Inherited carrier universe

The source-first carrier theorem supplies the complete carrier-semantic quotient $\mathfrak{U}_v$, including source-coherent parameter points, pointwise activity reduction, spectator stripping, and open symbolic family domains. Candidate identity is fixed before the present survival analysis:

$$x = [u, \theta] \in \mathfrak{U}_v, \quad \theta \in \Theta_u^{\text{part}}. \quad (3)$$

PCR-4 may select a parameter point as surviving or failing, but it may not change the active carrier trace assigned to that point.

2.4 Target object

Definition 2.1 (PCR-4 target). *The PCR-4 target is*

$$T_v^{(4)} = (T_v^{\text{osc}}, \mathcal{R}_v^{\text{osc}}, \mathfrak{S}_v^{\text{adm}}, \mathfrak{U}_v, \text{OneComp}, \text{EvalScope}_4), \quad (4)$$

where *OneComp* requires every successful coordinate to belong to one physical completion and *EvalScope₄* fixes the admitted source, interface, consistency, and extension evaluators.

2.5 Comparison class

Definition 2.2 (PCR-4 comparison class). *A candidate evaluation lies in $\mathfrak{C}_v^{(4)}$ iff it preserves: (i) the PCR-1 target and role packet; (ii) the PCR-2 target-facing trace and interface grammar; (iii) the PCR-3 source ontology, activity semantics, parameter point, and carrier equivalence; (iv) the charged-current boundary; (v) the one-completion requirement; and (vi) the admitted point-local or explicitly activated extended operator category.*

Theorem 2.3 (Import lock). *No downstream PCR-4 failure can alter $\mathcal{R}_v^{\text{osc}}$, $\mathfrak{S}_v^{\text{adm}}$, $\mathfrak{U}_v$, the PCR-3 activity reduction, or the PCR-2 comparison profile while remaining the same evaluation problem.*

Proof. Each object is an anchor of $T_v^{(4)}$ by definitions 2.1 and 2.2. Altering the role packet changes the target obligation; altering the interface changes the target-facing questions; altering the candidate universe or activity rule changes the object being evaluated. Such a move is an upstream correction or target/comparison-class update, not a repair of the candidate currently under evaluation. This is the fixed-domain consequence of reference and irreversibility in the AASC kernel [14, 18]. $\square$

2.6 Scope changes and premise revision

A genuine defect is classified by the mathematical object it changes. Exactly one of the following dispositions applies:

$$\text{candidate failure, PCR-1 reopening, PCR-2 reopening, PCR-3 reopening, new target fiber.} \quad (5)$$

Such a change revises an upstream premise or target fiber; it is not a modification of the same fixed evaluation problem.

Nonclaim 2.4 (No universal physics closure). *The target-relative lock does not assert that no richer experiment, operator category, curved background, nonlinear operational theory, or ultraviolet completion can be formulated. Such additions define explicit extensions of $T_v^{(4)}$.*

3 Dependency structure, source roles, and primary-physics boundary

3.1 Dependency stack

PCR-4 is downstream of a deliberately layered theorem architecture:

$$\begin{aligned} \text{AASC kernel} &\longrightarrow \text{Realized Physical Interior} \longrightarrow \text{SM carrier/symmetry closure} \\ &\longrightarrow \text{PCR-1--PCR-3} \longrightarrow \text{PCR-4}. \end{aligned} \quad (6)$$

The kernel supplies fixed-domain identity, standing, irreversibility, no-repair, and no-selector discipline. The Realized Physical Interior supplies the physical roles of local continuation, field carriage, localized persistence, gauge structure, deformation response, vacuum, time and causality. The Standard Model carrier and symmetry results fix the known gauge, representation, charge, anomaly, vacuum, global-form, and generalized-symmetry structures. PCR-1–PCR-3 then fix the role, interface, and candidate universe actually used here [13, 14, 18–21].

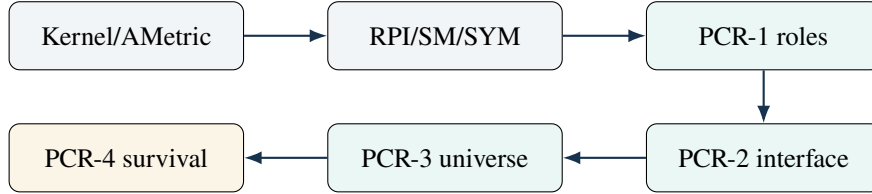


Figure 1: Logical dependency order. Each stage freezes a distinct burden before the next begins.

3.2 Source-role matrix

Table 1: Source roles in PCR-4.

Source	Permitted import	Blocked promotion
Kernel and AMetric	Fixed-domain identity, bivalence, no repair, no selector, standing conservation.	No neutrino mechanism, carrier family, or empirical conclusion.
RPI	Locality, continuation, field/particle roles, gauge, vacuum, time, causality.	No Standard-Model representation or candidate-specific dynamics.
SM carrier arc	Known gauge/matter/charge/flavor architecture and carrier closure.	No assumption that existing SM content occupies the PCR-1 roles.
SYM-9/SYM-11	Global form, line sectors, generalized-symmetry relevance and target typing.	No universal non-invertible or higher-form requirement.
SR1	Shared-ledger pullback, pairwise insufficiency, no free factorization.	No direct PCR-4 family disposition.
PCR-1	Ordered role complex and empirical role certificate.	No microscopic interpretation.
PCR-2	Complete target interface and semantic evaluator grammar.	No candidate generation or survivor result.

Source	Permitted import	Blocked promotion
PCR-3	Complete source-first carrier universe and carrier identity.	No interface pass/fail information.
ROC	Required downstream eligibility, compatibility and assignment objects.	No promotion of a PCR-4 survivor to realized occupant.

3.3 Primary-source physics boundary

Standard quantum-channel language is used in the target-operational sense established by Kraus and Stinespring and developed in open-system theory [3, 10, 11, 25]. Anomaly terms are evaluated using the appropriate local/global obstruction language rather than one undifferentiated cancellation slogan [2, 29]. EFT matching and decoupling are treated as scale-transport operations, not as candidate generators [8, 28]. Generalized-symmetry data are included only in activated operator sectors [1, 4, 5, 7, 26].

3.4 Proof hierarchy

The scientific claims are established by the definitions, physical constructions, and proofs in the manuscript. Supplementary tables may summarize the resulting loci and provenance, but they do not substitute for anomaly calculations, common-cone arguments, candidate completeness, or survivor completeness.

Nonclaim 3.1 (No source replacement). *The crosswalk does not reconcile or strengthen an imported theorem beyond its own scope. Where PCR-4 requires a target-specific evaluation, the relevant PCR-4 theorem is proved locally from the frozen imported interface and carrier data.*

3.5 Load-bearing dependency order

The imported results do not enter as a flat list of trusted conclusions. Their order fixes what PCR-4 may use and what it must still prove. The constructional kernel and AMetric boundary determine fixed-domain identity, fail-closed status, and the prohibition on hidden selectors. The Realized Physical Interior supplies the physical roles of localization, transport, carrier mediation, gauge stability, deformation response, vacuum support, temporal continuation, and causal dependence. PCR-1 then specializes those roles to the neutrino target and proves the ordered residual obligation. PCR-2 fixes the complete target-facing evaluator algebra and exact physical interface quotient. PCR-3 fixes the source-coherent carrier universe, pointwise source activity, and carrier-semantic identity. Only after all of these are frozen may PCR-4 ask which candidate instances satisfy the entire interface simultaneously.

The dependency order can be written schematically as

$$\text{Kernel} \longrightarrow \text{RPI} \longrightarrow \text{PCR1} \longrightarrow \text{PCR2} \longrightarrow \text{PCR3} \longrightarrow \text{PCR4}. \quad (7)$$

No arrow may be reversed without an explicit versioned reopening. In particular, a PCR-4 failure cannot be used to add a PCR-2 interface coordinate, regenerate the PCR-3 carrier universe, or reinterpret the PCR-1 role packet while continuing to claim the same theorem object.

Theorem 3.2 (Dependency-preserving evaluation). *Let two complete research states agree on every object upstream of PCR-4 but differ in a downstream PCR-5 assignment, phenomenological ranking, or expected realized occupant. Then their PCR-4 candidate domain, active constraint family, total evaluation vector, shared-witness pullback, and survivor universe are identical up to carrier-semantic equivalence.*

Proof. The PCR-4 evaluator receives only the frozen target, complete interface, source-first carrier instance, source parameter point, and physical witness data needed by the declared constraints. No formation rule contains an argument for a later assignment orbit, preferred model, posterior weight, or realized label. Therefore changing those downstream objects leaves the evaluator unchanged. Candidate identity and

parameter activity are already fixed pointwise by PCR-3, while target-facing evaluator completeness is already fixed by PCR-2. The common-cone condition depends only on the physical witness diagram of the current candidate. Hence all PCR-4 outputs remain invariant. $\square$

3.6 Primary-source versus corpus-source use

Ordinary quantum-channel, anomaly, effective-field-theory, and generalized-symmetry facts are cited to their primary mathematical or physical literature. The AASC-specific target, role, interface, carrier, and occupancy claims are imported from the preceding corpus papers. This distinction prevents a source-role ambiguity: a standard theorem can justify a channel representation or anomaly calculation without independently proving the AASC role hierarchy, while an AASC theorem can type a physical burden without replacing the underlying field-theoretic calculation.

3.7 No evidential inheritance by repetition

A statement does not become stronger because it appears in multiple companion papers. Every imported claim retains the status of its authoritative source theorem. The present paper therefore uses exact source pointers and theorem roles rather than treating the literature as a mutually reinforcing vote. If an upstream theorem is conditional, PCR-4 imports its condition. If a physical source establishes only a local or perturbative result, PCR-4 does not silently globalize it.

Dependency-failure diagnostic

An objection to PCR-4 has standing at the dependency level only if it identifies an upstream theorem used outside its stated scope, a missing imported hypothesis, a circular use of PCR-4 output, or a faithful same-target counterexample to one of the frozen interfaces. Merely preferring another explanatory order does not alter the proof.

4 Semantic survivor class

4.1 Why the semantic prior is required

Survival is defined semantically rather than by enumeration. The class is therefore meaningful independently of any tabulation, naming convention, or later family disposition.

Definition 4.1 (Semantic PCR-4 survivor). *A carrier architecture X belongs to $\text{CarrierSurv}_{\text{sem}}(T_v^{(4)})$ iff:*

1. $X \in \text{CarrierReal}_{\text{sem}}(T_v^{(3)})$ and is represented by its currently active, source-coherent PCR-3 carrier core;
2. X realizes the complete PCR-2 interface through the frozen charged-current boundary;
3. all target-admitted interface, source, quantum-consistency, vacuum, deformation, EFT, activated-extension, and simultaneous-realization evaluations are defined and standing-positive;
4. every successful evaluation factors through one physical witness architecture;
5. no failure is repaired by changing candidate identity or target after evaluation.

Definition 4.2 (Semantic survivor equivalence). *For $X, Y \in \text{CarrierSurv}_{\text{sem}}(T_v^{(4)})$, set*

$$X \equiv_{\text{surv}} Y \iff X \equiv_{\text{car}} Y.$$

The survivor quotient therefore inherits PCR-3 carrier identity rather than PCR-2 interface identity.

Proposition 4.3 (Survival is carrier-equivalence invariant). *If $X \equiv_{\text{car}} Y$, then $X \in \text{CarrierSurv}_{\text{sem}}(T_v^{(4)})$ iff $Y \in \text{CarrierSurv}_{\text{sem}}(T_v^{(4)})$.*

Proof. PCR-3 carrier equivalence preserves the complete active carrier trace, primitive parameter values modulo lawful reparameterization, source-coherence data, target attachments, and active incidence. Every PCR-4 evaluator is required to factor through these data together with the PCR-2 target-facing interface profile. Hence the definedness and value of every active evaluator, and the existence of the common witness cone, are preserved. $\square$

4.2 Survival trace

To make the semantic burden explicit, define the survivor trace

$$\tau_4(X) = (\chi_{\text{car}}(X), \Pi_\nu(X), \mathcal{J}_\nu^{\text{sim}}(X), \text{SrcStratum}(X), \text{ActiveExt}(X)). \quad (8)$$

A semantic survivor evaluator is any lawful, redescription-invariant, standing-bearing functional of τ_4 admitted by $T_\nu^{(4)}$. The trace is fixed before candidate disposition and contains no survivor label.

The tuple τ_4 is an expository abbreviation for the carrier, interface, extension, stratum, and co-instantiation entries used by the semantic definition. The complete survival-facing trace, semantic evaluator class, and factorization theorem are derived independently in section 5; no finite PCR-4 constraint list is used to define them.

Corollary 4.4 (Enumeration independence). *Membership in $\text{CarrierSurv}_{\text{sem}}(T_\nu^{(4)})$ is meaningful independently of any enumeration, catalogue, or tabular representation.*

Nonclaim 4.5 (No population theorem). *The semantic survivor class can be empty, multiply occupied, or singleton. Its definition does not imply any of those outcomes.*

4.3 Semantic survival as one-object physical realization

The semantic definition is intentionally stronger than a conjunction of independent existential statements. It is not enough that there exists some channel realizing the oscillation record, some representation assignment closing the gauge ledger, some topological sector canceling an anomaly, and some vacuum supporting the resulting fields. Those existential witnesses may belong to mutually incompatible constructions. A semantic survivor requires one physical architecture W_X whose restrictions and projections provide every active witness.

For clarity, let

$$\mathcal{D}_4(X) = (D_{\text{src}}, D_{\text{role}}, D_{\text{CC}}, D_{\text{chan}}, D_{\text{resp}}, D_{\text{loc}}, D_{\text{env}}, D_{\text{gauge}}, D_{\text{anom}}, D_{\text{vac}}, D_{\text{EFT}}, D_{\text{ext}}) \quad (9)$$

be the physical witness diagram associated with X . A semantic survivor is required to admit a cone

$$W_X \longrightarrow D_j \quad (j \in J_4) \quad (10)$$

whose restrictions agree on every shared overlap. The apex W_X is not a new primitive ontology. It is a witness that the various scientific descriptions concern one and the same carrier architecture.

Lemma 4.6 (Semantic survival is stricter than componentwise satisfiability). *There exist componentwise packets for which every individual active constraint has a positive witness but no semantic survivor exists.*

Proof. Take a propagation construction using active charge assignment q , an anomaly calculation closing only for a physically distinct charge assignment q' , and a vacuum branch supporting only the q' representation. Suppose each local calculation is valid in its own domain but no lawful carrier-equivalence map identifies q with q' in one candidate. Then each component constraint has a positive witness, but the overlap maps cannot commute and no common physical cone exists. Hence the packet is not a semantic survivor. $\square$

4.4 Role completeness inside the semantic definition

The semantic class preserves the strict nesting $r_\Delta \succ r_{\triangleright\triangleleft}$. The same physical channel and charged-current boundary must witness both obligations. A candidate cannot use an active-transfer structure to satisfy $r_{\triangleright\triangleleft}$ and an unrelated response mechanism to satisfy r_Δ unless both are physically co-instantiated in the same active carrier architecture. Conversely, one mechanism may satisfy both; the definition does not demand two carriers.

4.5 Target-relative rather than world-relative identity

A raw possible world may contain distant laboratories, unrelated hidden sectors, decoupled topological theories, and other carrier-capable objects. PCR-3 removes those structures unless they currently participate in the target-relative carrier core. PCR-4 inherits that reduction. The semantic survivor class therefore classifies the physical completion of the frozen target, not the total ontology of the world in which the completion is embedded.

Proposition 4.7 (Decoupled-world invariance). *Let Z be a source-coherent sector with no active target-relative carrier incidence, no target-facing interface witness, and no active PCR-4 constraint relation. Then*

$$X \in \text{CarrierSurv}_{\text{sem}}(T_v^{(4)}) \iff X \times Z \in \text{CarrierSurv}_{\text{sem}}(T_v^{(4)})$$

when both sides are interpreted after PCR-3 target-relative carrier reduction.

Proof. PCR-3 removes Z from the current carrier core, so the two active carrier traces are equal. All target-facing evaluators and overlap maps factor through that trace. The semantic survivor clauses therefore have identical truth values. $\square$

4.6 Falsifiability of the semantic prior

The semantic definition does not immunize the theorem from counterexample. A direct falsifier is a same-target physical carrier architecture that satisfies every scientific survival condition through one coherent witness system but is omitted from the canonical survivor set. The coverage theorem later proves that this cannot happen if PCR-3 candidate completeness and the PCR-4 evaluator are correct. Conversely, a canonical pass with no semantic physical interpretation falsifies the soundness direction.

5 Survival-consistency theory and evaluator completeness

5.1 Why relations must be fixed before the constraint list

A complete survival trace identifies every target-facing physical datum on which simultaneous survival may depend, but it does not by itself state the equations among those data. PCR-4 therefore fixes the semantic relation class before it introduces any publication-facing constraint ID. The construction uses only the frozen PCR-1 role burden, the PCR-2 target-facing interface, the PCR-3 active carrier ontology and source theory, the known-substrate consistency laws, and the one-completion requirement. Candidate names, family dispositions, and expected survivors are unavailable at this stage.

5.2 Raw survival presentations and the survival-facing trace

Definition 5.1 (Raw PCR-4 presentation). *A raw PCR-4 presentation is a tuple*

$$R = (x, \tilde{\mathfrak{F}}_v(x), \mathcal{D}_4(x)),$$

where $x \in \mathfrak{U}_v$ is a frozen PCR-3 carrier instance, $\tilde{\mathfrak{F}}_v(x)$ is its raw PCR-2 interface extraction, and $\mathcal{D}_4(x)$ is the typed physical witness diagram connecting source, role, boundary, channel, response, continuation, substrate, vacuum, EFT, extension, and co-instantiation data. The class of such presentations in one fixed target fiber is denoted $\mathcal{R}_4^{\text{raw}}$.

Definition 5.2 (Complete survival-facing trace). *Let*

$$\mathcal{T}_4 = \mathcal{T}_{\text{car}} \times \mathcal{T}_{\text{int}} \times \mathcal{T}_{\text{sub}} \times \mathcal{T}_{\text{ext}} \times \mathcal{T}_{\text{sim}} \times \mathcal{T}_{\text{str}}$$

be the product of:

1. the complete PCR-3 active carrier trace, source-coherence data, primitive parameter point, and source-activity stratum;
2. the complete PCR-2 target-facing trace of the raw interface extraction;
3. the known-substrate gauge, representation, charge, anomaly, vacuum, deformation, and matching data on which target consistency is permitted to depend;
4. every target- or candidate-activated global-form, extended-operator, defect, inflow, or topological datum;
5. the physical joint-incidence diagram, including witness identity, overlap maps, required commutative diagrams, and common-cone data;
6. the source and target stratum identifiers needed to distinguish a same-target failure from a target update.

The survival-facing trace map

$$\tau_4^{\text{surv}} : \mathcal{R}_4^{\text{raw}} \longrightarrow \mathcal{T}_4^\perp$$

returns these restrictions, or $\perp$ when the raw presentation is not sufficiently typed to form them.

The trace is fixed independently of all PCR-4 constraint IDs. It is scope-complete rather than encyclopedic: a new primitive witness sort not present in the frozen target or source ontology reopens the comparison class rather than being silently added to the same survival problem.

5.3 The many-sorted survival signature

Definition 5.3 (Survival signature). *The survival signature $\Sigma_4^{\text{surv}} = (\mathfrak{D}_4, \mathfrak{M}_4)$ has the physical sorts*

$$\mathfrak{D}_4 = \{C_{\text{car}}, R_{\text{role}}, B_{\text{CC}}, E_{\text{chan}}, P_{\text{rec}}, C_{\text{cont}}, S_{\text{sub}}, X_{\text{ext}}, J_{\text{sim}}, S_{\text{str}}\}.$$

The morphism families $\mathfrak{M}_4$ are the source interpretation and interface-extraction maps, charged-current attachments, preparation/effect maps, channel-to-record and channel-to-mode evaluations, support/dilation reductions, representation/charge/anomaly maps, vacuum restrictions, EFT matching maps, activated extension/inflow maps, and physical overlap maps required by one-completion semantics.

The sorts are not independently chosen projections. They are overlapping views of one candidate architecture. Their overlap maps and the physical equations among them are therefore part of the scientific object.

5.4 Semantic relation class before finite normal form

Definition 5.4 (Semantic survival relation). *A semantic survival relation is a lawful subobject*

$$Q \hookrightarrow \text{Im}(\tau_4^{\text{surv}})$$

with all of the following properties:

1. it is same-target and invariant under lawful source and interface redescription;
2. it is standing-bearing for whether one source-coherent carrier architecture realizes the frozen PCR-2 interface as one physical completion;
3. its arguments are existing sorts and morphisms of Σ_4^{surv} , rather than a new primitive witness type;

4. it is licensed by the frozen PCR-1–PCR-3 theorem stack, the known-substrate physical theory, or the constitutive theory of a target-activated extension sector;
5. it does not depend on model name, popularity, naturalness, likelihood ranking, expected-survivor identity, or any downstream PCR-5 assignment.

The class of all such relations is denoted $\text{Rel}_4^{\text{sem}}$. This definition is made before the named bundle $\mathcal{K}_4$.

For $Q \in \text{Rel}_4^{\text{sem}}$, let $\text{Supp}(Q)$ be its minimal collection of physical sorts and morphisms. The support may have arbitrary finite arity; no pairwise-factorization assumption is made.

Theorem 5.5 (Relation-support normal form). *Every $Q \in \text{Rel}_4^{\text{sem}}$ has one or more of the following support types:*

- R1.** source/interface identity;
- R2.** role/channel/response incidence;
- R3.** channel/record naturality;
- R4.** continuation/channel reduction;
- R5.** gauge/boundary attachment;
- R6.** representation/charge/anomaly identity;
- R7.** vacuum/deformation co-definition;
- R8.** EFT/matching naturality;
- R9.** extension/attachment coherence;
- R10.** one-completion co-instantiation and common-cone compatibility.

Conversely, every lawful relation of one of these support types that is licensed by the frozen source/target theory is a semantic survival relation. The ten types classify relation support; they are not a finite enumeration of individual equations.

Proof. Take a minimal support for Q . If it lies within one imported physical sort, Q is an intrinsic constitutive or admissibility relation of that sort and is recorded in the corresponding R1–R9 type. If its support meets more than one sort, form the connected support hypergraph whose vertices are primitive responsibilities and whose hyperedges are the morphisms on which Q depends. Every edge either identifies source and interface content, connects role to channel/response, computes records from a channel, reduces continuation to a channel, attaches the electroweak boundary, identifies representation/charge/anomaly data, co-defines vacuum/deformation data, transports data through EFT matching, attaches an activated extension, or asserts co-instantiation of the complete diagram. These are R1–R10. A higher-arity relation occupies all applicable support types and is retained as one irreducible hyperrelation; it is not replaced by pairwise marginals. Conversely, each licensed relation in those support classes is same-target, redescription invariant, and standing-bearing by the upstream theory that supplies it. No additional support type can be formed without a new primitive sort, morphism, boundary datum, or comparison regime. $\square$

Definition 5.6 (Survival-consistency theory). *The survival-consistency theory is*

$$\mathbb{S}_4^{\text{surv}} = (\mathfrak{D}_4, \mathfrak{M}_4, \mathfrak{E}_4),$$

where $\mathfrak{E}_4$ contains all relations in $\text{Rel}_4^{\text{sem}}$, organized by the support normal form of theorem 5.5, together with their lawful closure under substitution, restriction, composition, transport, matching, finite conjunction, and logical consequence. In particular, $\mathfrak{E}_4$ contains intrinsic equations, arbitrary finite-arity cross-signature relations, and one-completion/common-cone relations.

This is the independent scientific relation theory. The named PCR-4 predicates introduced later are a finite family of *relation schemas*; they are not a claim that finitely many stored Boolean outputs exhaust the theory.

Definition 5.7 (Global semantic survival locus). *The global semantic survival locus is*

$$\mathcal{A}_4^{\text{sem}} = \text{Mod}(\mathbb{S}_4^{\text{surv}}) \cap \text{Im}(\tau_4^{\text{surv}}).$$

Thus a trace is semantically admissible exactly when every intrinsic and cross-signature relation of the frozen survival theory holds and the complete witness diagram is physically co-instantiated.

5.5 Semantic uses and survival evaluators

Definition 5.8 (Semantic survival use). *A semantic survival use is a totalized, same-target, lawful-redescription-invariant map*

$$U : \mathcal{R}_4^{\text{raw}} \longrightarrow W_U^\perp$$

that is extensional in τ_4^{surv} , standing-bearing for simultaneous survival, and independent of model-name, likelihood, naturalness, popularity, or expected-survivor authority.

Lemma 5.9 (Survival-trace descent). *Every semantic survival use descends uniquely to a lawful map on $\text{Im}(\tau_4^{\text{surv}})$.*

Proof. Equal trace values are physically indistinguishable at the frozen survival interface by extensionality. Define the descended value on a trace by choosing any representative. The choice is independent of the representative, and uniqueness follows from surjectivity onto the trace image. $\square$

Definition 5.10 (Semantic survival evaluator). *A semantic survival evaluator is the membership test for a lawful subobject in $\text{Rel}_4^{\text{sem}}$, or for any subobject obtained from those relations by target-authorized lawful formation. The complete class is denoted $\text{PhysSurvEval}_4^{\text{sem}}$. No derivation from a named PCR-4 constraint ID is required by this definition.*

Theorem 5.11 (Survival-relation exhaustion). *Let Q be a same-target, standing-bearing physical consistency predicate on raw PCR-4 presentations. Exactly one of the following holds:*

- (i) $Q \in \text{Rel}_4^{\text{sem}}$ and hence is represented in $\mathbb{S}_4^{\text{surv}}$;
- (ii) Q is a concrete relation on existing frozen sorts that satisfies definition 5.4 but is absent from the asserted relation theory, and therefore is a direct falsifier of PCR-4 relation completeness;
- (iii) Q requires a new primitive witness sort, operator category, boundary datum, or comparison regime and is a target/source update.

Proof. If Q is extensional in the complete trace and uses only existing frozen sorts, inspect its minimal support. By theorem 5.5, its support is intrinsic or belongs to one or more of R1–R10. If it is licensed by the frozen physical theory, it is semantic and is included in $\mathbb{E}_4$ by definition. If such a licensed relation were missing from the theory presentation, it would satisfy clause (ii) and falsify the completeness claim rather than being silently treated as a pass. If Q requires a new sort or regime, clause (iii) holds. The alternatives are disjoint by fixed target typing and exhaustive by trace completeness and the support theorem. $\square$

5.6 Finite constraint normal form

For publication and family evaluation, the theory is presented by three generator layers:

1. intrinsic generator schemas $K_{\text{src}}, \dots, K_{\text{ext}}, K_{\text{no-repair}}, K_{\text{equiv}}$;

2. a cross-signature relation schema K_{rel} , whose typed residual packet

$$\mathbf{R}_4 = (R_{\text{src/int}}, R_{\text{role/chan}}, R_{\text{chan/rec}}, R_{\text{cont/chan}}, R_{\text{gauge/CC}}, R_{\text{rep/anom}}, R_{\text{vac}}, R_{\text{EFT}}, R_{\text{ext}})$$

contains every active relation in R1–R9, including irreducible higher-arity relations;

3. the one-completion/common-cone schema K_{sim} , containing R10.

Here $\mathbf{R}_4 = 0$ means that every active relation in the corresponding semantic relation families is satisfied; no finiteness, scalar-valuedness, or linearity is implied.

Definition 5.12 (Saturated constraint grammar). *Let K_4^{sat} be the target-authorized lawful closure of the intrinsic schemas, K_{rel} , and K_{sim} . The schema K_{rel} ranges over all active members of $\text{Rel}_4^{\text{sem}}$ with cross-signature support; it is not a single precomputed Boolean test.*

Theorem 5.13 (Semantic survival-evaluator generation). *The finite support-normal-form schemas generate the independent semantic survival evaluator class:*

$$\text{PhysSurvEval}_4^{\text{sem}} = K_4^{\text{sat}}.$$

Proof. Every intrinsic schema, every instance of K_{rel} , and every instance of K_{sim} is a semantic relation, so every normal-form consequence is semantic. Conversely, take any semantic evaluator. Its defining relation has an R1–R10 support normal form by theorem 5.5. Intrinsic instances occur in the corresponding intrinsic schema, cross-signature instances occur in K_{rel} , and one-completion instances occur in K_{sim} . Target-authorized formation then places the evaluator in K_4^{sat} . $\square$

Theorem 5.14 (Global constraint normal-form completeness). *For a same-target raw presentation R ,*

$$R \text{ passes every semantic survival evaluator} \iff \tau_4^{\text{surv}}(R) \in \mathcal{A}_4^{\text{sem}}.$$

Equivalently, a failure is an intrinsic inadmissibility, a failed cross-signature relation, failure of the common physical cone, an upstream omission, or a target/source update.

Proof. A model of $\mathbb{S}_4^{\text{surv}}$ satisfies every semantic relation and every lawful consequence. Conversely, each relation in $\mathfrak{C}_4$ defines a semantic evaluator. Passing all semantic evaluators therefore implies satisfaction of every relation of $\mathbb{S}_4^{\text{surv}}$, hence membership in $\mathcal{A}_4^{\text{sem}}$. The argument uses the global relation theory and arbitrary finite-arity relation schemas, not a product of marginal admissible loci. $\square$

Proposition 5.15 (Cross-relation mutation). *There exist raw same-target packets whose component signatures all lie in their marginal admissible subobjects and whose witness data admit a common apex object, but which fail a semantic cross-signature relation. Such packets are absent from $\mathcal{A}_4^{\text{sem}}$.*

Proof. Take a packet with one source architecture, event channel, representation/anomaly ledger, vacuum branch, and matching map. Keep every marginal object admitted and keep a common apex, but replace the EFT-to-interface comparison map by a map that does not commute with channel-to-record evaluation. The same objects still coexist, but the naturality square

$$R \circ \Pi_v \circ M = R \circ \tilde{\mathfrak{F}}_v$$

fails for some frozen record evaluator R . Thus all marginal tests and apex existence can pass while an R8 relation fails. The global theory rejects the packet through K_{rel} . $\square$

Corollary 5.16 (No product-of-marginals circularity). *PCR-4 completeness is derived from the independent semantic relation class, its support normal form, and the global model theory $\mathbb{S}_4^{\text{surv}}$. It does not follow from*

the number of named constraints, coordinatewise admissibility, common-cone existence alone, or the absence of known counterexamples.

6 Candidate evaluation

6.1 Disposition type

Definition 6.1 (Constraint disposition). *For a candidate $x \in \mathfrak{U}_v$ and constraint K , the disposition is*

$$\text{Disp}_4(x, K) \in \{PASS, FAIL, N/A, TARGET\ UPDATE\}. \quad (11)$$

N/A is permitted only when the corresponding target witness or operator category is theoremtically absent. TARGET UPDATE means that evaluation requires a different target fiber. Undefinedness is an evaluation defect, not a fifth scientific disposition.

Definition 6.2 (Candidate evaluation map). *The complete evaluation map is*

$$\text{Eval}_4 : \mathfrak{U}_v \longrightarrow \prod_{K \in \mathcal{K}_4} \{PASS, FAIL, N/A, TARGET\ UPDATE\}, \quad (12)$$

with each coordinate carrying its physical witness and formation data.

6.2 Totality

Theorem 6.3 (Evaluation totality). *Every $x \in \mathfrak{U}_v$ receives exactly one disposition for every PCR-4 constraint.*

Proof. PCR-3 supplies a source-coherent active carrier instance with fixed target attachments and parameter point. The complete survival-facing trace of definition 5.2 either exists or returns $\perp$. When it exists, every intrinsic or cross-signature relation schema of $\mathbb{S}_4^{\text{surv}}$ is either active, theoremtically absent, or requires a new target/source datum. An active relation is true or false on the fixed trace; a theoremtically absent relation yields N/A; a relation requiring a new primitive sort or regime yields TARGET UPDATE. Theorems 5.11 and 5.13 make these cases exhaustive without reducing survival to marginal coordinate membership. $\square$

6.3 Evaluation congruence

Theorem 6.4 (Carrier-equivalence congruence). *If $x \equiv_{\text{car}} y$, then*

$$\text{Eval}_4(x) = \text{Eval}_4(y) \quad (13)$$

and the corresponding physical witness packets are isomorphic.

Proof. The complete carrier trace and source parameter values agree up to lawful reparameterization. PCR-2 interface evaluation is invariant under its lawful target-semantic quotient, and the joint-incidence profile is physical rather than metadata-based. Every constraint therefore has the same definedness, output and co-instantiation structure on the two representatives. $\square$

6.4 Evaluation independence

The disposition is a function only of the carrier trace, source parameter point, target interface, and fixed evaluator definitions. It is independent of:

$$\begin{array}{lll} \text{expected survivor ID,} & \text{preferred model name,} & \text{PCR-5 assignment,} \\ \text{publication narrative,} & \text{empirical ranking.} & \end{array} \quad (14)$$

This independence is a consequence of the semantic factorization above, not of an implementation convention.

Proposition 6.5 (No ranking promotion). *Likelihood, naturalness, simplicity, model popularity, and numerical proximity cannot change $\text{Disp}_4(x, K)$ unless the corresponding criterion is itself part of the frozen target with an admitted physical witness.*

Proof. Such quantities are not among the formation rules of the PCR-2 evaluator grammar or the PCR-3 carrier trace. Introducing them as standing-bearing selectors adds a new target evaluator and therefore changes the target rather than refining the same constraint. $\square$

6.5 Evaluation as a typed partial-to-total completion

Many physical predicates begin life as partial functions: an anomaly polynomial is not defined before a representation and global-form fiber are fixed; a defect-fusion condition is irrelevant when no defect sector is active; an EFT matching test is undefined when source and target scales have not been declared. PCR-4 totalizes this situation by separating activity, applicability, target typing, and Boolean physical value.

For every constraint K , define an activity predicate $\text{Active}_K(x)$ before evaluating the predicate itself. The total evaluator is then

$$\text{Eval}_4(x, K) = \begin{cases} \text{N/A}, & \text{Active}_K(x) = 0 \text{ and absence is target-permitted,} \\ \text{TARGET UPDATE}, & \text{the evaluation requires a new target datum or operator category,} \\ \text{PASS}, & \text{Active}_K(x) = 1 \text{ and the complete predicate holds,} \\ \text{FAIL}, & \text{Active}_K(x) = 1 \text{ and the complete predicate fails.} \end{cases} \quad (15)$$

An active but scientifically unresolved predicate remains an open mathematical burden; it cannot be silently mapped to PASS.

Theorem 6.6 (No inapplicability laundering). *If K is active at x , no later knowledge of $\text{Eval}_4(x, K)$ may change $\text{Active}_K(x)$ while preserving the same target and candidate identity.*

Proof. Activity is a function of the frozen target signature and the PCR-3 pointwise active carrier trace. The value of K is computed only after those objects are fixed. Reclassifying an active locus as inactive because the predicate failed reverses the dependency order. A physical change that removes the locus changes the candidate point or target fiber. $\square$

6.6 Exactness over symbolic parameter domains

The evaluator is defined on instances, not family labels. If a family has source parameter space Θ_u^{part} , then each constraint induces a disposition stratification

$$\Theta_u^{\text{part}} = \Theta_{u,K}^{\text{PASS}} \sqcup \Theta_{u,K}^{\text{FAIL}} \sqcup \Theta_{u,K}^{\text{N/A}} \sqcup \Theta_{u,K}^{\text{TARGET UPDATE}}. \quad (16)$$

A complete PCR-4 manuscript must characterize these sets exactly enough to establish the final survivor locus. This may require algebraic geometry, representation theory, operator-domain analysis, topological classification, certified numerics, or combinations thereof. The theorem does not require every locus to have a simple closed form; it requires the reported locus to be complete.

6.7 Carrier-equivalence invariance

Evaluator congruence is stronger than invariance under model renaming. It requires invariance under every lawful carrier redescription, including source-normal-form changes, basis transformations, gauge representative changes, equivalent dilation representations, and lawful reparameterizations. Thus the evaluator descends from presentations to physical candidate classes.

Corollary 6.7 (Blind-evaluation invariance). *Any implementation that obtains different dispositions after permuting candidate order, replacing model names, or substituting carrier-equivalent source presentations is not an implementation of the PCR-4 evaluator.*

Corollary 6.8 (Family-map realization of totality). *For every family F and point $\theta \in \Theta_F^{\text{part}}$, the coordinate and relation values of $\widehat{\Phi}_F(\theta)$ determine the complete PCR-4 disposition. No summary table or family label may replace a missing coordinate by an expected pass.*

6.8 Diagnostic quantities and standing-relevant output

A physical evaluation may produce continuous diagnostics: anomaly residuals, unitarity defects, matching errors, spectral distances, or confidence intervals. Such quantities are retained for transparency, but the standing-relevant output is the typed disposition determined by the declared theorem. The boundary is not a ranking over near-misses. A candidate with a small nonzero gauge anomaly does not acquire partial survival.

7 Common physical witness structure

Definition 7.1 (Candidate witness packet). *For $x \in \mathfrak{U}_v$, define*

$$\mathcal{W}_4(x) = \left(\begin{array}{c} W_{\text{src}}, W_{\text{role}}, W_{\text{CC}}, W_{\text{chan}}, W_{\text{resp}}, W_{\text{loc}}, W_{\text{env}}, \\ W_{\text{gauge}}, W_{\text{rep}}, W_{\text{charge}}, W_{\text{anom}}, W_{\text{vac}}, W_{\text{def}}, W_{\text{EFT}}, W_{\text{ext}}, W_{\text{rel}}, W_{\text{sim}} \end{array} \right)_x. \quad (17)$$

7.1 Coordinate meanings

Table 2: PCR-4 witness packet.

Entry	Physical content	Survival question
W_{src}	PCR-3 active source diagram, coherence equations, parameter stratum.	Is the evaluated object the frozen candidate instance?
W_{role}	Capability evidence for $r_{\bowtie}$ and r_{Δ} .	Does one architecture carry the full nested burden?
W_{CC}	Operational charged-current preparation and detection boundary.	Are source and detection semantics preserved?
W_{chan}	Event-complete propagation instrument including null/loss outcomes.	Is active transfer distinguished from disappearance?
W_{resp}	Target-certified visible response-mode packet.	Are all required response classes reproduced by one completion?
W_{loc}	Local support, continuation and causal attachment.	Is propagation physically connected and non-pathless?
W_{env}	Medium/environment state, memory and coupling.	Is environmental participation source-defined and physical?
W_{gauge}	Electroweak gauge attachment and quotient.	Is the weak boundary preserved or explicitly enlarged?
W_{rep}	Active representation data.	Does the same active content carry propagation and consistency data?
W_{charge}	Charge witnesses and normalization class.	Are charge assignments common-ledger compatible?
W_{anom}	Local, global, mixed and activated generalized anomaly data.	Is the active quantum architecture consistent?
W_{vac}	Non-null vacuum/support baseline.	Does one physical baseline support the candidate?
W_{def}	Deformation response and parameter continuation.	Does the claimed survivor locus persist at fixed candidate identity?

Entry	Physical content	Survival question
W_{EFT}	Scale, matching, decoupling and coarse-graining maps.	Does the source architecture reach the target lawfully?
W_{ext}	Activated global-form, higher-form, defect, inflow or non-invertible structure.	Is extended structure mandatory, active, inert or absent?
W_{rel}	Cross-signature naturality, attachment, matching, co-definition, and inflow relations.	Do the admitted component values satisfy the global survival-consistency theory?
W_{sim}	Physical witness identity, overlap maps and common cone.	Do all positive claims belong to one construction?

7.2 Overlap structure

The packet is not a list. It is a typed diagram equipped with the relation theory $\mathbb{S}_4^{\text{surv}}$. In particular, the representation entering W_{chan} must be the representation whose charges and anomaly coefficients occur in W_{charge} and W_{anom} ; a hidden dilation used to produce loss must be the dilation recorded in W_{src} ; an inflow sector must be physically attached to the anomaly it cancels; the environmental state must belong to the vacuum and deformation realization used elsewhere.

Definition 7.2 (Physical common cone). *A physical common cone for $\mathcal{W}_4(x)$ is a witness object W_x with lawful maps into every active coordinate such that all required overlap diagrams commute and every coordinate is recovered by restriction or projection from W_x .*

Proposition 7.3 (Metadata non-authority). *File hashes, row IDs, theorem labels, and provenance paths do not contribute to common-cone existence.*

Proof. They can identify evidence records but do not alter the physical maps, witness identities or commutativity relations comprising the packet. If changing a metadata field changes survival, the metadata has been promoted to an illicit selector. $\square$

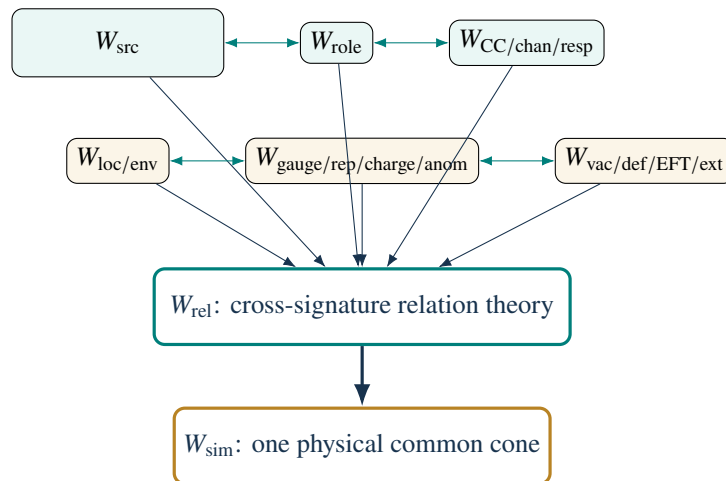


Figure 2: The candidate witness packet is a physical overlap diagram governed by cross-signature relations. Marginal admissibility and common-cone existence are both necessary; neither replaces the relation theory.

7.3 Physical identity in the witness diagram

The witness packet records more than values. It records which physical object carries each value. For example, a representation table is not merely a list of labels; it is attached to the active states that enter the

propagation channel. An anomaly calculation must restrict from that same representation object. A vacuum branch must support those same states. A matching map must descend from the same source architecture. These identity relations are encoded by the overlap morphisms of the witness diagram, while W_{rel} records the additional target/source equations that those overlapping values must satisfy. Common-cone existence establishes co-instantiation; it does not replace the cross-signature relation test.

Definition 7.4 (Witness-overlap graph). *The witness-overlap graph $\text{Ov}_4(x)$ has one vertex for every active witness coordinate and an edge or hyperedge for every required physical identity, restriction, transport, matching, or co-instantiation relation among them. Edge labels are physical maps, not database joins.*

The graph is promoted to a typed diagram by including composition and commutativity requirements. A common physical cone is an apex object whose projections reproduce all vertices and edges. This formalizes the phrase “the same candidate” across different physical descriptions.

7.4 Minimal overlap requirements

At minimum the diagram must enforce:

- O1.** channel and response use the same charged-current boundary and event instrument;
- O2.** propagation, representation, charge, and anomaly data refer to the same active carrier sectors;
- O3.** anomaly inflow and generalized-symmetry data live in the same global-form/extended-operator fiber;
- O4.** vacuum and deformation calculations support the same active source architecture;
- O5.** environmental and hidden-sector data used in the channel match those used in locality and EFT calculations;
- O6.** source-to-target matching commutes with the PCR-2 interface projection;
- O7.** every candidate witness agrees with the PCR-3 pointwise activity-reduced carrier trace.

Theorem 7.5 (Witness-packet sufficiency for simultaneous typing). *If all component predicates pass, the complete relation packet W_{rel} satisfies $\mathfrak{C}_4$, and the witness diagram has a common physical cone, then the component claims are typed as mutually consistent statements about one physical carrier architecture.*

Proof. The apex object provides one source-coherent physical carrier architecture together with projections to every component witness. Required commutativity identifies all shared carrier, boundary, environment, representation, vacuum, and matching data. Therefore no component success relies on a physically distinct auxiliary construction. This establishes co-instantiation and relation coherence; the completeness of the finite normal form is handled by theorems 5.13 and 5.14. $\square$

7.5 No metadata promotion

A tabular or computational representation may assign labels to the objects in this diagram for reproducibility, but such labels have no physical standing. What matters is the isomorphism class of the witness architecture and the commutativity of its physical overlap maps.

7.6 Witness packet as PCR-5 evidence

The shared-witness object later supports role compatibility, but PCR-4 does not infer that support automatically. PCR-5 must define which witness relations count as eligibility for each forced role and which combinations yield complete assignments. The PCR-4 packet is necessary input, not a precomputed assignment.

8 Simultaneous constraint normal form

8.1 Constraint family

Definition 8.1 (PCR-4 constraint bundle).

$$\mathcal{K}_4 = \left\{ \begin{array}{c} K_{\text{src}}, K_{\text{target}}, K_{\text{role}}, K_{\text{CC}}, K_{\text{chan}}, K_{\text{resp}}, K_{\text{loc}}, K_{\text{env}}, \\ K_{\text{gauge}}, K_{\text{rep}}, K_{\text{charge}}, K_{\text{anom}}, K_{\text{vac}}, K_{\text{def}}, K_{\text{EFT}}, K_{\text{ext}}, \\ K_{\text{rel}}, K_{\text{sim}}, K_{\text{no-repair}}, K_{\text{equiv}} \end{array} \right\}. \quad (18)$$

The displayed IDs are finite normal-form names for the relation families of $\mathbb{S}_4^{\text{surv}}$. The intrinsic IDs test single-sort admissibility, K_{rel} tests the cross-signature relation packet $\mathbf{R}_4$, and K_{sim} tests one-completion/common-cone existence. Let

$$\mathcal{K}_4^{\text{sat}} = \langle \mathcal{K}_4 \rangle_{\text{lawful}}$$

denote closure under target-authorized restriction, substitution, transport, pullback, relational composition, matching, and finite logical consequence. Fine-tuning, aesthetic economy, and post-hoc empirical preference are not formation rules.

Definition 8.2 (Active constraints). *For $x \in \mathfrak{U}_v$, $\mathcal{K}_4^{\text{act}}(x)$ contains every relation schema whose input witness sorts are present in the frozen target and candidate. An extended-sector relation is active only when the target or candidate activates its operator category. Cross-relations are active whenever all sorts in their support are present.*

Definition 8.3 (Survivor predicate).

$$\text{Surv}_4(x) = 1 \iff \text{Pull}_4(x) = 1, \quad \tau_4^{\text{surv}}(R_x) \in \mathcal{A}_4^{\text{sem}}, \quad \text{Disp}_4(x, K) = \text{PASS} \quad \forall K \in \mathcal{K}_4^{\text{act}}(x), \quad (19)$$

with every N/A theorem-certified and no TARGET UPDATE disposition.

The middle condition is not an unexplained extra test: by theorem 5.14 it is equivalent to passage of the saturated normal form generated by $\mathcal{K}_4$.

8.2 Constraint completeness

Theorem 8.4 (Constraint-bundle completeness). *Every same-target standing-bearing reason for a carrier instance to fail simultaneous survival is represented by one intrinsic relation family, one cross-relation in K_{rel} , or failure of the common-cone/typing conditions.*

Proof. By theorem 5.11, a same-target survival relation is either a consequence of the independently fixed theory $\mathbb{S}_4^{\text{surv}}$, a concrete omitted same-target relation that falsifies its completeness, or a target update. By theorem 5.13, the lawful closure of the intrinsic IDs, K_{rel} , and K_{sim} generates every consequence of $\mathbb{S}_4^{\text{surv}}$. Hence every same-target failure is detected by this normal form unless it is an explicit falsifier of the relation theory. $\square$

Theorem 8.5 (Global-relation mutation theorem). *There exist raw presentations whose marginal intrinsic signatures lie in their individual admissible loci and whose witness diagram admits a common apex object, but which fail a same-target cross-coordinate relation. Every such presentation fails K_{rel} and is excluded from $\mathfrak{S}_v^{(4)}$.*

Proof. Use the mutation of proposition 5.15: keep one source architecture, admitted component values, and a common apex, but replace one EFT-to-interface comparison map by a map that fails naturality with channel-to-record evaluation. The component objects coexist, yet the resulting trace is not a model of $\mathbb{S}_4^{\text{surv}}$, so K_{rel} fails. $\square$

Theorem 8.6 (Fail-closedness). *If $\text{Disp}_4(x, K) = \text{FAIL}$ for any active load-bearing K , or if $\tau_4^{\text{surv}}(R_x) \notin \mathcal{A}_4^{\text{sem}}$, then $x \notin \mathfrak{S}_v^{(4)}$. A lawful physical modification yields another PCR-3 instance x' or a target update; it cannot repair x as the same act.*

Proof. The failed relation belongs to the standing-bearing target/interface/source theory. Any modification that changes its value changes the active carrier trace, source parameter point, interface extraction, overlap relation, or target fiber and therefore names a different PCR-3 instance or a target update. A modification changing none of these leaves the failed semantic relation unchanged. $\square$

Corollary 8.7 (No soft survival). *Approximate proximity to a constraint surface is not partial survival unless an explicit physical tolerance is part of the frozen target.*

8.3 Failure decomposition

For a nonsurviving point, it is useful to distinguish the first load-bearing failure in the fixed dependency order from the complete set of failed intrinsic and cross-signature relations. This ordering is diagnostic only and assigns no physical priority to an earlier-listed condition.

8.4 Why the bundle is complete but not universal

Constraint completeness is target-relative. PCR-2 fixes the complete target-facing evaluator/interface grammar; PCR-3 fixes the complete active carrier trace and source-coherence theory; the known substrate contributes its gauge, representation, charge, anomaly, vacuum, deformation, and matching equations; one-completion semantics contributes the cross-relation and common-cone families. A purported new survival condition must therefore be an intrinsic relation, a cross-relation, an omitted primitive datum or relation, or a target/source update.

This does not mean that every possible ultraviolet consistency condition is silently assumed. If a candidate introduces a new operator category, spacetime regime, or source sort, the corresponding relation theory is activated only when it belongs to the frozen target/candidate comparison class; otherwise the candidate receives a target-update disposition.

Theorem 8.8 (Constraint-factorization quadrichotomy). *Let Q be a same-target standing-bearing relation proposed as necessary for PCR-4 survival. Then exactly one of the following holds:*

- (i) Q is intrinsic and factors through one coordinate family in $\mathcal{K}_4$;
- (ii) Q is cross-coordinate and factors through K_{rel} and the relevant overlap morphisms;
- (iii) Q exposes an omitted target evaluator, carrier distinction, substrate relation, or overlap morphism and therefore falsifies an upstream completeness theorem or $\mathbb{S}_4^{\text{surv}}$;
- (iv) Q requires a new primitive target datum, operator category, or comparison regime and defines a target/source update.

Proof. Apply theorem 5.11 to the minimal support of Q . The normal-form conclusion follows from theorems 5.13 and 5.14. $\square$

8.5 Constraint interaction

The bundle is not a product of independent marginal predicates. Passing K_{gauge} can change the domain on which K_{anom} is evaluated; vacuum reduction can change active representation content; EFT matching can activate or remove extended sectors; environmental coupling can alter the event-complete channel. Those intrinsic changes are evaluated on one fixed candidate point, while K_{rel} enforces the equations relating their values and K_{sim} enforces physical co-instantiation.

8.6 No optimization principle

The bundle defines a feasible set, not an objective function. There is no built-in preference for fewer fields, lower dimension, smaller coupling, local over nonlocal carriers, particle over defect descriptions, or conventional over novel mechanisms. Any later minimality theorem must be stated as a separate physical removal result.

9 Raw, co-instantiated, and survivor moduli

9.1 Raw evaluation object

Let $\mathcal{J}_4$ index the active evaluation projections. For each j , let $M_{4,j}$ be the projection-level space after quotienting only presentation skin. The raw object is

$$\mathcal{M}_4^{\text{raw}} = \prod_{j \in \mathcal{J}_4} M_{4,j}. \quad (20)$$

It contains all tuples obtainable by independently selecting projection data; it is not a space of physical completions.

9.2 Pullback and survivor objects

Definition 9.1 (Shared-witness pullback).

$$\mathcal{M}_4^{\text{pull}} = \{m \in \mathcal{M}_4^{\text{raw}} : \exists W \text{ Cone}_4(W, m)\}. \quad (21)$$

Definition 9.2 (Simultaneous survivor object).

$$\mathcal{M}_4^{\text{surv}} = \{m \in \mathcal{M}_4^{\text{pull}} : K(m) = 1 \ \forall K \in \mathcal{K}_4^{\text{act}}(m)\}. \quad (22)$$

Thus

$$\boxed{\mathcal{M}_4^{\text{surv}} \hookrightarrow \mathcal{M}_4^{\text{pull}} \hookrightarrow \mathcal{M}_4^{\text{raw}}.} \quad (23)$$

Theorem 9.3 (Pairwise compatibility is insufficient). *There exist raw packets for which every pair of active projections admits a pairwise witness system while no single physical common cone exists. Hence pairwise or checklist compatibility does not imply membership in $\mathcal{M}_4^{\text{pull}}$.*

Proof. Take three projections: propagation/response P , anomaly/representation A , and vacuum/environment V . Let P be realized by active representation R_1 and environment state ω_1 . Let A close only for representation R_2 , with a lawful pairwise map from R_1 to R_2 after forgetting the channel action. Let V be compatible with P through ω_1 and with A through a different state ω_2 , while no single state/representation pair realizes all three overlap maps. Each pair has a compatible witness system, but the triple diagram has no cone. This is the standard obstruction to inferring global compatibility from pairwise intersections and is the neutrino-facing analogue of SR1's witness-split construction. $\square$

Theorem 9.4 (Shared-ledger necessity). *If $x \in \mathfrak{S}_V^{(4)}$, then its evaluation packet lies in $\mathcal{M}_4^{\text{pull}}$.*

Proof. Survival includes one-completion semantics and K_{sim} . Every successful coordinate must therefore descend from one candidate-specific physical witness system, which is precisely the common-cone condition. $\square$

Corollary 9.5 (No free factorization). *A load-bearing change in any survivor projection must be promoted to a complete PCR-3 candidate instance and re-evaluated through the whole packet. It is freely variable only if it is lawful skin.*

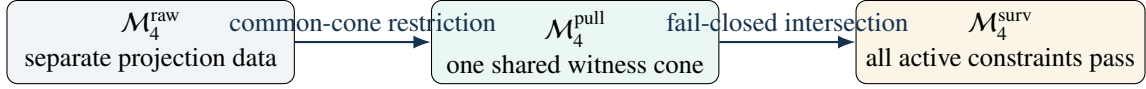


Figure 3: Formal PCR-4 collapse. Componentwise success lies in the raw product; physical co-instantiation defines the pullback; simultaneous survival is the fail-closed subobject.

9.3 Relational interpretation of the pullback

The notation $\mathcal{M}_4^{\text{pull}}$ need not presuppose a sophisticated ambient category. It can be read extensionally as the subset of raw component packets for which one candidate-specific witness object realizes all component projections and overlaps. When the relevant witness categories are available, it is a genuine limit/pullback object. In either reading, the scientific content is identical: independent existential witnesses are insufficient.

Theorem 9.6 (Nondegenerate collapse). *Suppose the raw comparison regime contains at least one packet whose component predicates are individually satisfiable but whose physical overlap diagram lacks a common cone, or at least one common-cone packet failing an active constraint. Then*

$$\mathcal{M}_4^{\text{surv}} \subsetneq \mathcal{M}_4^{\text{raw}}.$$

Proof. The stated packet lies in $\mathcal{M}_4^{\text{raw}}$ by componentwise satisfiability. In the first case it is absent from $\mathcal{M}_4^{\text{pull}}$; in the second it is present in $\mathcal{M}_4^{\text{pull}}$ but absent from $\mathcal{M}_4^{\text{surv}}$. Hence it is absent from the survivor object, proving proper inclusion. $\square$

9.4 Failure of a hidden Helly principle

There is no licensed theorem asserting that pairwise common witnesses imply a global common witness. Such a principle would itself be a nontrivial completion theorem and would need independent physical proof for the relevant witness category. PCR-4 therefore treats global co-instantiation as an explicit condition.

9.5 Relation to fiber products over the target trace

The raw component spaces are all target-relative. Their pullback is not taken over an empty formal point but over shared target and carrier data: charged-current boundary, active carrier trace, environment, representation, global form, vacuum branch, and scale/matching structure. This makes witness splitting detectable even when component output values agree numerically.

9.6 Survivor object as a scientific residue

The survivor moduli object is the exact residue after common-witness and fail-closed reduction. It may still contain many carrier classes and continuous strata. PCR-4 makes no claim that the residue is discrete, finite, or small. Its theorem is about exact correspondence, not cardinality.

10 Nested role, charged-current boundary, channel, and response constraints

10.1 Role capability as one-architecture burden

Definition 10.1 (Role-capability predicate). $\text{RoleCap}_4(x) = 1$ iff the same carrier architecture supplies active charged-current sector nonalignment r_{pc} , the full visible response multiplicity r_{Δ} , and the PCR-1 nesting relation $r_{\Delta} \succ r_{\text{pc}}$.

Theorem 10.2 (Nested-role survival). *If $x \in \mathfrak{S}_v^{(4)}$, then $\text{RoleCap}_4(x) = 1$. A candidate that provides visible flavor transfer but only one lawful response class, or provides multiple modes in a sector unattached to the charged-current boundary, fails.*

Proof. The PCR-1 role packet is an immutable target coordinate. PCR-2 fixes the boundary and visible-mode interface. Survival requires the same completion to realize both coordinates through the common witness

packet. Separate mechanisms in incompatible candidate worlds do not satisfy K_{sim} . $\square$

10.2 Charged-current boundary

Let $\{P_\alpha\}$ be the operational charged-current source/detection projectors and ρ_α a source state supported in sector α . A survivor preserves their operational meaning and attaches its propagation architecture to those same sectors.

Theorem 10.3 (Boundary preservation). *A candidate that changes the preparation or detection semantics used to define the PCR-1 record packet receives TARGET UPDATE, not PASS, unless the change is a lawful representative redescription preserving the target-facing trace.*

Proof. The boundary is part of the frozen target. A physical change in what counts as source flavor, detection event, or null outcome changes the empirical record map and therefore defines a new target fiber. $\square$

10.3 Event-complete channel

The operational propagation object is an event-complete instrument

$$\mathcal{E}_\lambda = \{\mathcal{E}_{\lambda,o}\}_{o \in \Omega_{\text{act}} \sqcup \Omega_{\text{null}}}, \quad \sum_o \mathcal{E}_{\lambda,o} \text{ trace preserving}, \quad (24)$$

which includes active charged-current outcomes and null/loss events. This formulation covers unitary propagation, open-system dynamics, invisible decay, absorption, and enlarged-state-space dilations at the target-operational level [3, 10, 11, 23, 25].

Proposition 10.4 (Loss is not active transfer). *Invisible loss or decay alone does not occupy r_{pass} . A survivor must satisfy*

$$\exists \alpha \neq \beta, \lambda : \quad \text{Tr}[P_\beta \mathcal{E}_\lambda(\rho_\alpha)] > 0. \quad (25)$$

10.4 Visible response multiplicity

The candidate's projected response must contain the frozen target mode classes:

$$\mathfrak{M}_{\text{vis}}(T_v^{\text{osc}}) \hookrightarrow \mathfrak{M}_{\text{vis}}(x) \quad (26)$$

under the same lawful calibration and environment transports used by PCR-1/PCR-2. Candidate-specific redefinition of the target modes is not permitted.

Theorem 10.5 (Response completeness). *If $x \in \mathfrak{S}_v^{(4)}$, then one event-complete channel/attachment architecture reproduces every target-certified visible mode class and preserves the active transfer incidence.*

Proof. This is the conjunction of K_{chan} , K_{resp} , K_{CC} , and the common-cone condition. The empirical source certificates are imported rather than refitted inside PCR-4 [6, 9, 24]. $\square$

Nonclaim 10.6 (No microscopic interpretation). *The channel and response constraints do not imply masses, mixing matrices, a particular Hilbert-space dimension, or a unique microscopic propagation law.*

10.5 Channel-native formulation across microscopic realizations

The event-complete instrument is deliberately operational. A unitary three-state model, a nonunitary reduced dynamics, an explicit dilation, a decay model, a medium-coupled process, or a defect-supported realization can all induce such an instrument on the frozen source/detection target. The survival analysis evaluates the induced physical interface without assuming that any one microscopic representation is fundamental.

The visible active transition probability is

$$P_{\alpha \rightarrow \beta}^x(\lambda) = \text{Tr}[P_\beta \mathcal{E}_{x,\lambda}(\rho_\alpha)], \quad \alpha \neq \beta, \quad (27)$$

with null/loss outcomes included in the complete normalization. The nonalignment role requires at least one nonzero active transition in the frozen target class. The response-multiplicity role requires the full target-certified operational mode packet after lawful nuisance quotienting.

Proposition 10.7 (Microscopic many-to-one freedom). *Distinct microscopic carriers can induce the same event-complete target instrument and the same visible response packet. PCR-4 therefore cannot infer carrier identity from channel equality.*

Proof. Stinespring dilations of a fixed channel are nonunique up to environment isometry, and physically distinct source architectures can also share the same reduced operational response. PCR-3 carrier identity retains the source-semantic distinctions, while PCR-2 interface identity quotients only target-facing equivalence. PCR-4 preserves both layers. $\square$

10.6 Environmental dependence of the response packet

A response mode is compared only after applying the frozen environment and calibration transports. A candidate may predict matter-dependent frequencies, attenuation, or phase structure, but the same environment witness must be used in the channel, response, and locality calculations. Candidate-specific reassignment of an environmental effect to a new target mode is not permitted.

10.7 No mass-language shortcut

A conventional mass/mixing model can be evaluated as a local-carrier candidate, but PCR-4 does not use mass eigenvalues or a PMNS matrix as definitions of r_Δ or r_{vac} . Such parameters enter only through their operational predictions and their full gauge/anomaly/vacuum/EFT carrier architecture. This maintains the carrier-neutral endpoint of PCR-1 and PCR-2.

11 Locality, transport, causality, and environmental participation

11.1 Transport object

PCR-2 imports a target-relative continuation structure connecting source, propagation and detection regions. The present analysis asks whether the candidate realizes that structure through its active carrier architecture rather than through pathless global coordination or detector-side retargeting.

Definition 11.1 (Lawful transport realization). *A candidate transport realization consists of local/support domains L_s, L_p, L_d , lawful continuation maps between them, a causal/support relation, and the candidate's environment/medium data, all descending from one witness system.*

Theorem 11.2 (Locality and continuation constraint). *If $x \in \mathfrak{S}_v^{(4)}$, then every target-visible propagation response is carried by a lawful continuation from the frozen source boundary to the frozen detection boundary. A purported effect requiring no admitted physical path or requiring undeclared global coordination fails K_{loc} unless the richer continuation structure is already part of the PCR-3 carrier trace.*

Proof. The Realized Physical Interior requires local continuation domains, bounded transport and witness-dependent causality for standing-bearing physical persistence [19]. PCR-2 specializes these roles to the neutrino interface. A candidate that lacks the corresponding support maps does not realize the same target; adding them after failure changes candidate identity. $\square$

11.2 Environmental and medium co-carriers

An environment can be part of the carrier architecture rather than an external nuisance. To survive, an environmental candidate must specify:

1. the physical environment state and its support;
2. the candidate-environment incidence and memory structure;

3. the common vacuum/deformation realization;
4. the reduced event-complete channel;
5. the lawful source-to-target matching map.

Proposition 11.3 (Environment is neither automatic carrier nor automatic skin). *An environment enters the carrier core exactly when its currently active incidence participates in the PCR-3 carrier trace. A merely external calibration field is interface/support data; a physically incident medium co-carrier is candidate structure.*

11.3 Causal discipline

Candidate nonlocality is typed, not dismissed by name. A nonlocal or memory-bearing construction survives only when its support and causal semantics are part of the frozen comparison class or an explicitly declared target extension. Operational no-signalling constraints are evaluated on the complete source-to-detection instrument, not on a hidden internal representation alone.

Example 11.4 (Two-environment witness split). *Suppose the response profile is produced using medium state ω_1 , while the vacuum/deformation and anomaly analysis is valid only for ω_2 , with no lawful state map preserving both. The candidate can pass separate environmental calculations yet fails W_{sim} and is outside $\mathcal{M}_4^{\text{pull}}$.*

11.4 Support categories and target locality

Candidates can realize transport through local fields, extended objects, environmental memory, hidden-sector dilations, or other source-authorized structures. The target does not force one support category, but it does force a lawful path from source preparation to detection record. Every active support component must be present in the PCR-3 carrier trace and in the shared witness diagram.

Theorem 11.5 (Transport/interface commutation). *For a PCR-4 survivor, the projection from the source carrier architecture to the PCR-2 target-facing interface commutes with lawful continuation from source to detector.*

Proof. The complete witness packet contains both the support/continuation maps and the target-interface projection. The common-cone overlap clauses require their commutativity. A noncommuting diagram would mean that the reported target response is not the restriction of the candidate's actual transport architecture, so K_{sim} or K_{loc} fails. $\square$

11.5 Environment as carrier rather than calibration

The same physical medium can appear in two roles. A known density profile used only to compute target propagation is part of the fixed environmental boundary. A proposed quantum or classical environment that stores memory, induces transition, or carries hidden information is a candidate co-carrier. PCR-4 distinguishes them by source activity and carrier trace, not by whether the environment is experimentally accessible.

11.6 No environment patchwork

If one experiment is explained with environment state E_1 and another with E_2 , the candidate must contain a source-defined physical rule relating the two to the corresponding experimental boundary conditions. Otherwise the environment has been used as an unrestricted selector. The rule itself is part of the candidate and must satisfy vacuum, locality, and EFT constraints.

12 Gauge attachment, representation, charge, and anomaly closure

12.1 Gauge attachment

The carrier must attach to the frozen electroweak/lepton substrate without silently replacing the charged-current boundary. The active representation used by the propagation architecture is part of the physical witness packet.

Definition 12.1 (Gauge-attachment packet).

$$\mathcal{G}_4(x) = (G_x, R_x, Q_x, \mathcal{A}_x, \iota_x), \quad (28)$$

where G_x is the active gauge/global-form datum, R_x the active representation content, Q_x the charge witness, $\mathcal{A}_x$ the relevant anomaly ledger, and ι_x the physical attachment to the known substrate.

Theorem 12.2 (Representation identity across projections). *If $x \in \mathfrak{S}_v^{(4)}$, the representation and charge content entering propagation, charged-current attachment, anomaly evaluation, vacuum support, and EFT matching are restrictions of the same active carrier architecture.*

Proof. This is the overlap condition between W_{chan} , W_{gauge} , W_{rep} , W_{charge} , W_{anom} , W_{vac} , and W_{EFT} . Choosing different representation ledgers for separate calculations yields a raw tuple without a common cone. $\square$

12.2 Anomaly bundle

For each active symmetry/background/operator category, PCR-4 evaluates the corresponding local, mixed, global or generalized obstruction. Standard perturbative anomaly conditions are represented by the appropriate traces or index polynomials; global anomalies require the faithful group and global data [2, 29].

Definition 12.3 (Anomaly-consistent candidate). *A candidate is anomaly-consistent iff every active gauge obstruction vanishes or is canceled by a physically incident, source-coherent mechanism contained in the same carrier trace, and every active global/'t Hooft anomaly satisfies the target's matching or inflow condition.*

Theorem 12.4 (No spectator anomaly laundering). *A disconnected spectator sector cannot cancel the anomaly of an active candidate. An anomaly-canceling sector contributes only when its current physical incidence places it in the same active carrier architecture.*

Proof. Anomaly cancellation is a statement about the complete active quantum measure/representation ledger. A disconnected sector has no physical incidence in that ledger under PCR-3 activity reduction. Adding an incidence activates a different candidate instance, which must be evaluated as such. $\square$

12.3 Global-form dependence

Changing the faithful global form can alter line operators, monopole sectors, theta periodicities and generalized symmetry data even when the local Lie algebra is unchanged [1, 7, 26]. PCR-4 therefore treats global form as active exactly when the candidate or target activates the corresponding extended operator category.

Nonclaim 12.5 (No complete Standard Model rederivation). *PCR-4 imports the frozen Standard Model carrier and symmetry ledgers. It does not rederive the full product group or charge lattice, nor does it infer numerical couplings from anomaly closure.*

12.4 Gauge consistency versus global anomaly matching

PCR-4 keeps distinct the failure of a gauge architecture and the nontrivial anomaly of a global or generalized symmetry. A noncanceled gauge anomaly is inconsistent with the candidate's quantum gauge realization. A global or higher-form anomaly can instead be a valid matching constraint, provided the candidate contains the declared infrared realization or inflow structure. Both calculations must use the same active representations, charges, global form, and extended sectors.

Theorem 12.6 (Anomaly witness locality in candidate space). *An anomaly cancellation or inflow witness can contribute to $K_{\text{anom}}(x)$ only if it belongs to the active carrier trace of x at the evaluated parameter point.*

Proof. PCR-3 source activity determines the current carrier core before PCR-4. The anomaly constraint is a physical predicate on that core and its active global/extended data. A disconnected or source-null sector has no physical overlap map into the common cone, so it cannot contribute. Activating a new coupling produces another candidate instance. $\square$

12.5 Charge normalization and lawful equivalence

Overall charge normalization can be presentation skin when it preserves the physical charge lattice and all anomaly coefficients after corresponding coupling normalization. Relative charge assignments, signs, discrete classes, and global-form data are physical when they alter the active witness ledger. PCR-4 therefore quotients only lawful normalization changes and never uses normalization freedom as a cancellation mechanism.

12.6 Gauge-vacuum-anomaly coupling

Vacuum reduction can change the residual gauge group, active spectrum, and anomaly ledger. PCR-4 does not compute these in separate targets. The candidate must supply one vacuum-mediated reduction map from the pre-vacuum active architecture to the residual target, and anomaly consistency must be preserved along that same map.

12.7 Global-form activation

When a candidate activates line or monopole sectors, local Lie-algebra anomaly checks are incomplete. PCR-4 imports the faithful group and extended-operator data from the candidate's target/global-form fiber. A result obtained in another fiber is not a lawful substitute.

13 Vacuum support, deformation strata, and EFT matching

13.1 Vacuum-support object

A candidate cannot borrow one vacuum to define its propagation channel and another incompatible vacuum to stabilize its carrier or cancel its anomaly. The vacuum packet contains the non-null baseline, candidate state/phase, boundary conditions, and active carrier incidence.

Theorem 13.1 (Vacuum compatibility). *If $x \in \mathfrak{S}_v^{(4)}$, there exists one source-coherent vacuum/support realization in which the active carrier architecture, propagation channel, gauge/anomaly ledger, environment and deformation maps are simultaneously defined.*

Proof. Vacuum support is a coordinate of the PCR-2 interface and a physical overlap in W_{sim} . Incompatible vacuum realizations fail the common-cone condition even when each projection is separately meaningful. $\square$

13.2 Parameter-stratified deformation

For a PCR-3 family u , define

$$\Theta_u^{\text{surv}} = \{\theta \in \Theta_u^{\text{part}} : \text{Surv}_4([u, \theta]) = 1\}. \quad (29)$$

The locus may be empty, discrete, continuous, disconnected, stratified, or defined by source equations and inequalities.

Theorem 13.2 (Exact survivor-stratum theorem). *For every family u , Θ_u^{surv} is invariant under lawful source reparameterization and is determined pointwise before any PCR-5 role assignment.*

Proof. Evaluation is carrier-equivalence congruent by theorem 6.4; lawful reparameterization preserves the PCR-3 active carrier trace. Therefore pass/fail dispositions and common-cone existence descend to the quotient parameter space. PCR-5 receives the resulting locus but has no authority to modify it. $\square$

13.3 EFT matching

A source architecture may be represented at the oscillation scale by an effective operator, reduced channel, nonlocal kernel or integrated-out sector. The matching packet must preserve the active carrier trace and target interface through lawful scale transport, following the standard EFT separation between source degrees of freedom and low-energy operators [8, 28].

Theorem 13.3 (No carrier creation by matching). *If the active source carrier architecture is absent at a parameter point, EFT matching cannot create its standing as an occupant. It may transmit the effect of an already present source architecture or define a law-only candidate already represented in PCR-3.*

Proof. Matching maps source data to a target-effective description. They do not alter PCR-3 source activity. A purported effective term with no source carrier is either a law-only source candidate or an undeclared source/target update; it is not a repair of an absent carrier. $\square$

13.4 Decoupling

Decoupling can make a carrier invisible at the target scale while leaving effective response data. The survival analysis evaluates both the source carrier identity and the matched target interface. It does not merge the source candidate with a carrier-distinct law-only candidate merely because their low-energy interfaces coincide.

13.5 Vacuum existence versus vacuum compatibility

The existence of some vacuum state is weaker than PCR-4 vacuum survival. The vacuum must support the active carrier architecture, preserve the declared source/detection and propagation semantics, admit the required deformation germ, and descend consistently to the target EFT. A candidate cannot pass by pairing a formal vacuum of one theory with the channel of another.

Definition 13.4 (Common vacuum witness). *A common vacuum witness for x is a baseline state or sector Ω_x together with maps showing that the active carrier, representation/charge, anomaly, environment, channel, and EFT objects are all defined over or reduced from the same branch.*

Proposition 13.5 (Vacuum scalar non-authority). *An additive scalar vacuum-energy offset or gauge-dependent order parameter cannot by itself certify K_{vac} . Only the quotient-stable physical vacuum/baseline structure and its common-witness relations are relevant.*

13.6 Deformation strata without naturalness

The deformation condition can be exact on a lower-dimensional or isolated stratum. PCR-4 does not reject such a locus for being fine-tuned unless the target theorem explicitly requires stability under a larger deformation class. Conversely, a candidate that works only after changing an anchor or target boundary fails the fixed-scope theorem even if the change is numerically small.

13.7 Matching preserves carrier responsibility

Integrating out a field can remove it from an effective description while retaining its effects in operators and anomaly terms. PCR-4 distinguishes effective presentation from carrier identity: PCR-3 retains the source carrier architecture, while K_{EFT} checks that the target interface is lawfully reproduced. The effective theory cannot be relabeled as a law-only candidate if its source carrier trace contains active ultraviolet structure.

Theorem 13.6 (No carrier creation by matching). *A matching or renormalization operation cannot create target-relative carrier standing that was absent from the source architecture. It may transport, encode, or hide an existing active carrier contribution.*

Proof. Matching is a map from a source theory to an effective target description. Its admissibility presupposes a source-coherent active architecture and a lawful projection. If the required role appears only after introducing

a new source object or attachment, that object belongs to another PCR-3 candidate. If no source carrier bears the effect, the matching coefficient is an unsupported generator rather than a transport of the same candidate. $\square$

14 Global form, higher-form, higher-group, defect, and non-invertible relevance

14.1 Relevance gate

Generalized-symmetry structure is neither universally imposed nor universally ignored. For each candidate define

$$\text{ExtStatus}(x) \in \{\text{mandatory}, \text{active}, \text{inert}, \text{absent}\}. \quad (30)$$

The status is determined by the frozen target operator category together with candidate-activated lines, surfaces, defects, junctions, backgrounds, global forms and anomaly/inflow data.

Definition 14.1 (Activated extension). *An extended sector is activated when a target-admitted evaluator or a source-coherent candidate incidence depends on its operator/defect data in a standing-bearing way.*

Theorem 14.2 (Relevance-gated generalized consistency). *If $\text{ExtStatus}(x) \in \{\text{mandatory}, \text{active}\}$, then the complete extended-sector fusion, junction, module, global-form, anomaly and attachment data enter $\mathcal{W}_4(x)$ and $\mathcal{K}_4^{\text{act}}(x)$. If the status is inert or absent, those data cannot be used to select or repair the candidate.*

Proof. PCR-2 proves that generalized-symmetry data enter the interface exactly when a target-facing witness activates them. PCR-3 includes source-coherent topological and defect families without privileging them. The common-cone condition then requires any active extended data to belong to the same physical architecture. Inert data change no target-semantic evaluator and are not standing-bearing. $\square$

14.2 Non-invertible candidates

A non-invertible carrier candidate must supply source-coherent defect objects, fusion multiplicities, junction spaces, associator/module data and physical attachment. PCR-3 already enforces constitutive coherence. PCR-4 adds target relevance, quantum consistency, vacuum/deformation support, propagation/response adequacy and shared realization. A defect-localized TQFT that changes no bulk target witness remains inert; one used for anomaly inflow or propagation response becomes active and is tested in full [4, 5].

Theorem 14.3 (No generalized-symmetry default). *The PCR-1 role complex alone does not force a higher-form, higher-group or non-invertible interface coordinate. Such structure is mandatory only after an independent target or candidate witness activates it.*

Proof. The point-local PCR-1/PCR-2 target can be represented by carrier architectures whose operational channel contains no extended operator datum. Therefore generalized structure is not entailed solely by $\mathcal{R}_V^{\text{osc}}$. Conversely, when a candidate requires such data, ignoring them would omit an active target evaluator. The relevance gate is therefore necessary in both directions. $\square$

Nonclaim 14.4 (No exclusion of generalized completions). *A core-inert classification does not prove that generalized symmetry is absent from a richer completion. It states only that the current target does not use it.*

14.3 Four-way relevance disposition

For every candidate, generalized-symmetry data receive one of four statuses:

$$\text{GenStat}(x) \in \{\text{mandatory}, \text{active-constraining}, \text{present-inert}, \text{absent}\}. \quad (31)$$

Mandatory means the frozen target itself requires the sector. Active-constraining means the candidate source trace activates it and some PCR-4 predicate depends on it. Present-inert means the source presentation

contains an extended sector that PCR-3 spectator/activity reduction shows to be target-inert. Absent means no such source object exists.

Theorem 14.5 (Relevance congruence). *Carrier-equivalent candidate instances have identical generalized-symmetry relevance status and equivalent active extended-sector witness diagrams.*

Proof. Carrier equivalence preserves the complete active carrier trace, including global form, genuine extended objects, target attachments, fusion/module data, and source activity. The relevance gate is a function of these data and the frozen target trace. Hence it descends to candidate classes. $\square$

14.4 Constitutive coherence versus target survival

Fusion associativity, module laws, defect-junction coherence, and topological source objecthood are PCR-3 source-coherence conditions. PCR-4 does not re-test them as target viability; it assumes the candidate is a genuine source object and then tests physical attachment, anomaly, vacuum, channel, and common-cone conditions. This preserves the distinction between mathematical objecthood and physical survival.

14.5 Extended-sector changes after failure

A topological or non-invertible sector introduced after a local candidate fails anomaly closure defines another source architecture. The source-first carrier universe already contains every lawful active hybrid, which is evaluated as a complete object together with the extended sector's own gauge, vacuum, matching, and channel conditions.

15 Explicit survival maps and family specialization

15.1 Why symbolic does not mean unevaluated

An exact survivor locus need not be a finite list or a closed-form numerical region. It may be algebraic, semialgebraic, operator-theoretic, categorical, or defined by the nonemptiness of a precisely specified moduli problem. What is not sufficient is the circular description “the points that pass all constraints.” This section replaces that placeholder by an explicit target-relative evaluation map for every PCR-3 family.

15.2 Universal candidate data

For a source family F and a participating parameter point $\theta \in \Theta_F^{\text{part}}$, let

$$X_F(\theta)$$

be the PCR-3 activity-reduced semantic carrier representative. The source interpretation determines:

1. charged-current attachment maps $A_{F,\theta}^s$ and $A_{F,\theta}^d$;
2. an event-complete instrument $\{\mathcal{I}_{\lambda,o}^{F,\theta}\}_{o \in \mathcal{O}_{\text{evt}}}$, or a target-fiber failure;
3. source and detector states/effects $\rho_\alpha^{F,\theta}$ and $P_\beta^{F,\theta}$;
4. the event kernels

$$K_{\beta\alpha}^{F,\theta}(\lambda) = \text{Tr} \left[P_\beta^{F,\theta} \mathcal{I}_\lambda^{F,\theta}(\rho_\alpha^{F,\theta}) \right]; \quad (32)$$

5. support, causal-continuation, environment, representation, charge, anomaly, vacuum, deformation, matching, extension, and physical-incidence data.

These objects are derived from the source interpretation; they are not separately chosen fitting ledgers.

15.3 Instrument and record residuals

For each event outcome o , let

$$J_{\lambda,o}^{F,\theta}$$

be the Choi operator of $\mathcal{I}_{\lambda,o}^{F,\theta}$. Define the instrument residual

$$\Phi_{\text{inst}}^F(\theta) = \left(\{J_{\lambda,o}^{F,\theta}\}_{\lambda,o}, \left\{ \sum_o (\mathcal{I}_{\lambda,o}^{F,\theta})^\dagger(\mathbf{1}) \right\}_\lambda \right). \quad (33)$$

Its admissible locus is

$$J_{\lambda,o}^{F,\theta} \succeq 0, \quad \sum_o (\mathcal{I}_{\lambda,o}^{F,\theta})^\dagger(\mathbf{1}) = \mathbf{1} \quad \text{for all admitted } \lambda, o. \quad (34)$$

The outcome set includes the null/loss event, so disappearance cannot be mistaken for active flavor transfer.

Let Rec_T be the frozen target family of record evaluators and let $\mathfrak{M}_{\text{vis}}(T)$ be the PCR-1/PCR-2 visible-mode packet. Define

$$\Phi_{\text{rec}}^F(\theta) = \left(R(K^{F,\theta}) \right)_{R \in \text{Rec}_T}, \quad (35)$$

$$\Phi_{\text{mode}}^F(\theta) = \mathfrak{M}_{\text{vis}}(X_F(\theta)). \quad (36)$$

The record/mode locus is

$$R(K^{F,\theta}) = R(T) \quad \forall R \in \text{Rec}_T, \quad \mathfrak{M}_{\text{vis}}(T) \hookrightarrow \Phi_{\text{mode}}^F(\theta) \quad (37)$$

through the frozen lawful mode-equivalence relation. The role coordinate $r_{\bowtie}$ additionally requires

$$\exists \alpha \neq \beta, \lambda : K_{\beta\alpha}^{F,\theta}(\lambda) > 0, \quad (38)$$

and r_Δ requires the complete target-visible mode packet. Thus loss alone fails (38).

15.4 Boundary and gauge attachment

Let $\rho_{F,\theta}$ be the active gauge representation and ρ_{CC} the frozen charged-current boundary representation. The attachment signature is

$$\Phi_{\text{CC}}^F(\theta) = \left([A_{F,\theta}^s], [A_{F,\theta}^d], [\rho_{F,\theta}] \right).$$

Its admissible locus requires

$$A_{F,\theta}^{s,d} \rho_{F,\theta}(g) = \rho_{\text{CC}}(g) A_{F,\theta}^{s,d} \quad \forall g \quad (39)$$

on the target-facing source/detection subspaces, preservation of the operational flavor labels, and preservation of residual electromagnetic charge. Equality only after a change of source/detection semantics is a target update, not a pass.

15.5 Transport, environment, and causal support

Let $\mathcal{D}_{\text{caus}}^F(\theta)$ be the moduli set of source-declared causal/local dilations or equivalent support witnesses for the event instrument. An element contains source and detector supports, protocol order, medium path data, and a projection to the effective instrument. The transport/environment locus is

$$\mathcal{D}_{\text{caus}}^F(\theta) \neq \emptyset, \quad \text{supp } \mathcal{I}_{\lambda,o}^{F,\theta} \subseteq C_\lambda^+ \quad (40)$$

in a local representative, or the corresponding target-authorized causal-dilation statement in a nonlocal effective representative. The environment map

$$\eta \longmapsto \mathcal{I}_{\lambda,\eta}^{F,\theta}$$

must be defined on the frozen environment domain and preserve the charged-current boundary. An undeclared source/detector interaction or pathless global coordinator lies outside the locus.

15.6 Representation, charge, and anomaly residual

Write the new active left-handed Weyl content of $X_F(\theta)$ as

$$\{(n_i, R_{3i}, R_{2i}, Y_i)\}_{i \in I_{F,\theta}},$$

with n_i the multiplicity and with every representation and charge taken from the same carrier trace used by the channel. Let $A_3(R)$ denote the cubic $SU(3)$ anomaly coefficient and $T_a(R)$ the quadratic index. Define

$$\mathbf{A}_F(\theta) = \begin{pmatrix} \sum_i n_i d(R_{2i}) A_3(R_{3i}) \\ \sum_i n_i d(R_{2i}) T_3(R_{3i}) Y_i \\ \sum_i n_i d(R_{3i}) T_2(R_{2i}) Y_i \\ \sum_i n_i d(R_{3i}) d(R_{2i}) Y_i^3 \\ \sum_i n_i d(R_{3i}) d(R_{2i}) Y_i \\ \omega_2(F, \theta) \end{pmatrix} + \mathbf{A}_{\text{inflow}}^F(\theta), \quad (41)$$

where ω_2 is the relevant global $SU(2)$ -parity obstruction and $\mathbf{A}_{\text{inflow}}^F$ is present only when an active source-coherent inflow/defect sector exists in the same witness diagram. Since the frozen Standard Model boundary is anomaly-free, the candidate contribution must satisfy

$$\boxed{\mathbf{A}_F(\theta) = 0} \quad (42)$$

together with every activated faithful-global-form or background anomaly condition. A formal Boolean label “anomaly free” without the common representation/charge witness is not an evaluation.

15.7 Vacuum and deformation moduli

Define $\mathcal{V}_F(\theta)$ to be the set of source-admitted baseline states or sectors Ω such that:

1. the candidate source equations or equivalent baseline conditions are satisfied on Ω ;
2. the frozen charged-current and residual electromagnetic boundary is preserved;
3. the active carrier, event instrument, representation/charge, anomaly, environment, and extension data are all defined on or reduced from the same branch;
4. every declared isolated, singular, or phase-boundary stratum is typed explicitly;
5. no vacuum datum is imported after a failure.

The vacuum/deformation locus is

$$\mathcal{V}_F(\theta) \neq \emptyset. \quad (43)$$

No open-neighborhood naturalness condition is imposed. The set may be isolated or stratified.

15.8 EFT and matching moduli

Let $\mathcal{M}_F(\theta)$ be the set of source-to-target matching diagrams

$$M_{\mu \rightarrow \Lambda} : X_F(\theta) \longrightarrow \tilde{\mathfrak{F}}_\nu(X_F(\theta))$$

that preserve the complete target-facing trace, active carrier responsibility, anomaly/inflow data, vacuum branch, visible-mode incidence, and activated extension witnesses. The EFT locus is

$$\mathcal{M}_F(\theta) \neq \emptyset. \quad (44)$$

15.9 Activated extension and simultaneous realization

Let $\mathcal{X}_F(\theta)$ be the source-coherent active extension packet. If its operator category is active, the fusion, module, global-form, inflow, and attachment data must satisfy their target-facing constitutive equations. If it is inactive, it contributes the distinguished inert value and cannot repair another failure.

Let $\text{Cone}_F(\theta)$ be the set of physical cones over the complete witness diagram whose restrictions reproduce all signatures above. The simultaneous-realization locus is

$$\text{Cone}_F(\theta) \neq \emptyset. \quad (45)$$

15.10 Cross-signature relation residuals

Marginally admitted objects and a common apex do not yet guarantee global survival consistency. Define the typed relation residual packet

$$\mathbf{R}_F(\theta) = \left(\begin{array}{c} R_{\text{src/int}}^F, R_{\text{role/chan}}^F, R_{\text{chan/rec}}^F, R_{\text{cont/chan}}^F, \\ R_{\text{gauge/CC}}^F, R_{\text{rep/anom}}^F, R_{\text{vac}}^F, R_{\text{EFT}}^F, R_{\text{ext}}^F \end{array} \right) (\theta). \quad (46)$$

The distinguished value 0 denotes satisfaction of the corresponding typed equality or commutative diagram; no linearity is implied. Its components are:

R1. $R_{\text{src/int}}^F = 0$ iff the raw interface, active carrier trace, source parameter point, and target attachment are restrictions of the same source-interpreted object $X_F(\theta)$.

R2. $R_{\text{role/chan}}^F = 0$ iff the carrier bearer assigned to r_{src} and r_{Δ} is the bearer entering $\mathcal{I}^{F,\theta}$, $K_{\beta\alpha}^{F,\theta}$, Φ_{mode}^F , and $A_{F,\theta}^{s,d}$.

R3. $R_{\text{chan/rec}}^F = 0$ iff

$$K_{\beta\alpha}^{F,\theta}(\lambda) = \text{Tr} \left[P_{\beta}^{F,\theta} \mathcal{I}_{\lambda}^{F,\theta} (\rho_{\alpha}^{F,\theta}) \right]$$

and every frozen record and visible-mode evaluator is applied to that same kernel.

R4. $R_{\text{cont/chan}}^F = 0$ iff an admitted continuation witness $(\iota_F, \mathcal{U}_F, \pi_F) \in \mathcal{D}_{\text{caus}}^F(\theta)$ satisfies

$$\mathcal{I}_{\lambda,o}^{F,\theta} = \pi_F \circ \mathcal{U}_{F,\lambda,o} \circ \iota_F$$

(or the corresponding lawful reduction in the declared process category), with the same environment state used in transport and vacuum data.

R5. $R_{\text{gauge/CC}}^F = 0$ iff the charged-current attachments satisfy (39) for the same active representation and residual charge data used by the channel.

R6. $R_{\text{rep/anom}}^F = 0$ iff $\mathbf{A}_F(\theta)$ is computed from that same representation/charge object and every inflow term is physically attached to the anomaly and operator fiber it cancels.

R7. $R_{\text{vac}}^F = 0$ iff the source equations, carrier nodes, channel, representation, environment, and extension packet are defined on or reduced from one branch $\Omega \in \mathcal{V}_F(\theta)$.

R8. $R_{\text{EFT}}^F = 0$ iff every $M \in \mathcal{M}_F(\theta)$ commutes with active-carrier reduction, interface extraction, record evaluation, visible-mode extraction, anomaly transport, vacuum restriction, and activated-extension transport.

R9. $R_{\text{ext}}^F = 0$ iff the active extension packet satisfies its fusion/module/junction/global-form equations and its attachment or inflow maps terminate in the same target/operator fiber as the effect they support.

These residuals are the family specialization of the independently fixed relation theory $\mathbb{S}_4^{\text{surv}}$. In particular, a packet can pass every marginal condition and possess a common apex while failing $\mathbf{R}_F = 0$.

15.11 Universal global evaluation map

Collect the intrinsic signatures into

$$\Phi_F(\theta) = \left(\Phi_{\text{src}}^F, \Phi_{\text{CC}}^F, \Phi_{\text{inst}}^F, \Phi_{\text{rec}}^F, \Phi_{\text{mode}}^F, \Phi_{\text{tr}}^F, \mathbf{A}_F, \mathcal{V}_F, \mathcal{M}_F, \mathcal{X}_F, \text{Cone}_F \right) (\theta). \quad (47)$$

The complete family evaluation map is

$$\widehat{\Phi}_F(\theta) = (\Phi_F(\theta), \mathbf{R}_F(\theta)). \quad (48)$$

Let $\mathcal{X}_4^{\text{int}}$ denote the typed locus satisfying equations (34), (37) to (40) and (42) to (45) together with source/target identity and activated-extension constitutive coherence. The family realization of the global semantic model locus is

$$\mathcal{X}_4^{\text{sem}} = \{(y, r) : y \in \mathcal{X}_4^{\text{int}}, r = 0\}. \quad (49)$$

It is not a Cartesian product of marginal pass loci.

Theorem 15.1 (Exact family-evaluation theorem). *For every PCR-3 family F ,*

$$\Theta_F^{\text{surv}} = \widehat{\Phi}_F^{-1}(\mathcal{X}_4^{\text{sem}}) \subseteq \Theta_F^{\text{part}}. \quad (50)$$

This is an exact global symbolic locus, not the circular comprehension “the points that pass all constraints.”

Proof. Every intrinsic component of Φ_F and every relation residual in $\mathbf{R}_F$ is obtained by source interpretation and the independently derived survival-consistency theory. Membership in $\mathcal{X}_4^{\text{sem}}$ states the intrinsic target equations, nonempty moduli requirements, all cross-signature relations, and common-cone existence. By theorem 5.14, this is equivalent to passing every semantic survival evaluator. Hence the preimage is exactly the survivor locus. $\square$

Corollary 15.2 (Failure set from the global family map). *For $\theta \notin \Theta_F^{\text{surv}}$, the complete failure set consists of:*

1. *every intrinsic coordinate outside its lawful locus;*
2. *every nonzero relation residual in $\mathbf{R}_F$;*
3. *the common-cone disposition;*
4. *any target/source-update disposition.*

This set is nonempty. A diagnostic first failure may be chosen only after the complete intrinsic and relational failure sets have been computed.

16 Evaluated survivor loci I: existing-substrate, law-only, and local carriers

16.1 Family overlaps and the meaning of a pure stratum

The PCR-3 source families are not disjoint labels. A carrier built only from the frozen substrate can also be law-bearing, composite, or environment-supported. To state actual survivor results, define the *pure stratum* of a family to be the locus in which no additional occupancy-bearing source coordinate from another principal family is active.

16.2 Strict existing-substrate stratum

Let Θ_{BASE}^0 be the PCR-3 locus whose active carrier trace is exactly the frozen massless-neutrino Standard Model carrier architecture, with no new law/process object, no derived collective node, no active environment co-carrier, and no additional hidden, boundary, topological, or defect incidence.

Theorem 16.1 (Strict existing-substrate exclusion).

$$\Theta_{\text{BASE}}^{0,\text{surv}} = \emptyset.$$

Every point of the strict base stratum fails K_{role} , K_{chan} , and K_{resp} .

Proof. PCR-1 proves that the frozen minimal renormalizable massless-neutrino Standard Model occupies neither r_{ps} nor r_{Δ} . Its propagation channel is flavor-block preserving on the charged-current boundary and carries no target-certified multi-mode response. Since the strict base stratum adds no law, incidence, derived node, or co-carrier capable of changing that channel, equations (37) and (38) fail at every point. $\square$

Theorem 16.2 (Existing-substrate reduction). *Every surviving existing-substrate candidate belongs to at least one of the overlap strata*

$$F_{\text{BASE}} \cap (F_{\text{LAW}} \cup F_{\text{COMP}} \cup F_{\text{ENV}} \cup F_{\text{HYB}}).$$

Hence

$$\Theta_{\text{BASE}}^{\text{surv}} = (\Theta_{\text{BASE} \cap \text{LAW}}^{\text{surv}} \cup \Theta_{\text{BASE} \cap \text{COMP}}^{\text{surv}} \cup \Theta_{\text{BASE} \cap \text{ENV}}^{\text{surv}} \cup \Theta_{\text{BASE} \cap \text{HYB}}^{\text{surv}}) / \simeq_{\text{inst}}.$$

Proof. PCR-3 proves that new occupancy-bearing content on the fixed primitive substrate must occur as a new process/law, a derived collective/composite organization, an active environment co-carrier, or a finite hybrid of those structures. If none occurs, the candidate lies in Θ_{BASE}^0 , which is excluded by theorem 16.1. $\square$

16.3 Law-only family

A law-only point is determined by a source-coherent process object L_{θ} on existing carrier nodes. Its exact survivor locus is

$$\begin{aligned} \Theta_{\text{LAW}}^{\text{surv}} &= \left\{ \theta \in \Theta_{\text{LAW}}^{\text{part}} : J_{\lambda,o}^{\theta} \succeq 0, \sum_o (I_{\lambda,o}^{\theta})^{\dagger}(\mathbf{1}) = \mathbf{1}, \right. \\ &\quad R(K^{\theta}) = R(T) \forall R \in \text{Rec}_T, \mathfrak{M}_{\text{vis}}(T) \hookrightarrow \mathfrak{M}_{\text{vis}}(X_{\theta}), \\ &\quad \exists \alpha \neq \beta, \lambda : K_{\beta\alpha}^{\theta}(\lambda) > 0, \\ &\quad L_{\theta} \text{ is gauge-covariant on } \mathcal{B}_{\text{CC}}, \mathcal{D}_{\text{caus}}^{\theta} \neq \emptyset, \\ &\quad \mathcal{V}_{\text{LAW}}(\theta) \neq \emptyset, \mathcal{M}_{\text{LAW}}(\theta) \neq \emptyset, \mathbf{R}_{\text{LAW}}(\theta) = 0, \text{Cone}_{\text{LAW}}(\theta) \neq \emptyset \left. \right\}. \end{aligned} \tag{51}$$

No new chiral source node is present, so the gauge-anomaly contribution remains the frozen Standard Model value provided the law object is gauge covariant.

Theorem 16.3 (Law-only locus is nonempty).

$$\Theta_{\text{LAW}}^{\text{surv}} \neq \emptyset.$$

Proof. A source-coherent witness is supplied by the gauge-invariant dimension-five lepton-number-violating operator on the existing lepton and Higgs fields,

$$\mathcal{L}_5 = \frac{1}{2} \kappa_{\alpha\beta} \left(L_{\alpha}^T C i \sigma_2 H \right) \left(H^T i \sigma_2 L_{\beta} \right) + \text{h.c.}$$

[27]. On the fixed electroweak vacuum it yields

$$M_{\nu} = \frac{v^2}{2} \kappa, \quad H_{\text{prop}}(E, \eta) = \frac{1}{2E} M_{\nu} M_{\nu}^{\dagger} + V_{\text{env}}(\eta).$$

Choose κ so that $M_{\nu} M_{\nu}^{\dagger}$ has the target-certified two independent visible response classes and does not commute with every charged-current flavor projector. The induced unitary propagation channel, supplemented

by the frozen production/detection instrument and null outcome, satisfies equations (34), (37) and (38). The operator is gauge invariant, adds no new chiral node or anomaly, is defined on the fixed vacuum, and is itself the low-energy EFT matching object. The complete witness diagram is one physical construction, and its channel-record, attachment, gauge, anomaly, vacuum, EFT, and overlap data satisfy the global relation theory $\mathbb{S}_4^{\text{surv}}$. Therefore at least one point lies in $\Theta_{\text{LAW}}^{\text{surv}}$. $\square$

Corollary 16.4 (No-new-primitive-carrier survival). *PCR-4 does not force enlargement of the primitive carrier-node inventory. At least one survivor uses only the frozen Standard Model field nodes together with a new law/operator object.*

16.4 New local primitive carriers

Let the active new local content be

$$\{(n_a, R_{3a}, R_{2a}, Y_a, \epsilon_a)\}_{a \in A}$$

with source-coherent local attachment operators $O_{\text{att},a}$, vacuum/mass data, and primitive parameter point θ . Define the local attachment moduli

$$\mathcal{A}_{\text{loc}}(\theta) = \{O_{\text{att},a} : O_{\text{att},a} \text{ is a gauge-invariant source operator linking } L, H \text{ to the active local nodes}\}.$$

The exact local survivor locus is

$$\begin{aligned} \Theta_{\text{LOCAL}}^{\text{surv}} = \{ \theta \in \Theta_{\text{LOCAL}}^{\text{part}} : & \mathcal{A}_{\text{loc}}(\theta) \neq \emptyset, \mathbf{A}_{\text{LOCAL}}(\theta) = 0, \\ & \text{equations (34), (37) and (38) hold,} \\ & \mathcal{D}_{\text{caus}}^{\text{LOCAL}}(\theta) \neq \emptyset, \mathcal{V}_{\text{LOCAL}}(\theta) \neq \emptyset, \\ & \mathcal{M}_{\text{LOCAL}}(\theta) \neq \emptyset, \mathbf{R}_{\text{LOCAL}}(\theta) = 0, \text{Cone}_{\text{LOCAL}}(\theta) \neq \emptyset \}. \end{aligned} \quad (52)$$

All representation, charge, anomaly, channel, and vacuum quantities in this formula are evaluated on the same active local content.

Theorem 16.5 (Local-carrier locus is nonempty).

$$\Theta_{\text{LOCAL}}^{\text{surv}} \neq \emptyset.$$

Proof. Introduce $n \geq 2$ gauge-singlet right-handed fermions

$$N_{Ra} \sim (\mathbf{1}, \mathbf{1})_0$$

and the source-coherent interactions

$$\mathcal{L}_{\text{att}} = -\bar{L}_\alpha (Y_\nu)_{\alpha a} \tilde{H} N_{Ra} - \frac{1}{2} \overline{N_{Ra}^c} (M_R)_{ab} N_{Rb} + \text{h.c.}$$

[22]. Gauge singlets contribute zero to every component of $\mathbf{A}_{\text{LOCAL}}$. Choose Y_ν and M_R so that the effective active-neutrino propagation kernel has nontrivial charged-current nonalignment and the two target-visible response classes. The fixed electroweak vacuum supports the attachment, and integrating out heavy singlets or retaining light ones gives a lawful EFT matching diagram. The visible and hidden sectors, anomaly ledger, vacuum, channel, and matching map belong to one source architecture, so the common cone exists. The resulting trace is a model of $\mathbb{S}_4^{\text{surv}}$, including the representation-charge-anomaly and vacuum-EFT cross-relations. Thus the local survivor locus is nonempty. $\square$

Corollary 16.6 (At least two carrier-distinct survivor classes). *The law-only witness of theorem 16.3 and the active local-singlet witness of theorem 16.5 are not carrier-equivalent: the second contains active primitive*

local nodes absent from the first. Hence

$$\left| \mathfrak{G}_\nu^{(4)} \right| \geq 2$$

at the carrier-semantic level.

16.5 Local anomaly equations are evaluated, not labelled

For a general local family point, $\mathbf{A}_{\text{LOCAL}}(\theta) = 0$ in equation (52) is the complete set of perturbative and activated global anomaly equations equations (41) and (42). Vectorlike pairs, gauge singlets, chiral spectra, and explicit inflow sectors are therefore treated by the same equation rather than by model-name expectations.

Table 3: Evaluated first-tranche results.

Stratum	Exact result	Supporting result
Strict existing substrate	Empty.	PCR-1 nonoccupancy; failure of active transfer and visible-mode conditions.
Existing substrate with new organization	Reduced exactly to law/composite/environment/hybrid overlaps.	PCR-3 source-family factorization and theorem 16.2.
Law-only	Exact locus equation (51); nonempty.	Dimension-five gauge-invariant operator witness.
New local	Exact locus equation (52); nonempty.	Gauge-singlet local-fermion witness with common anomaly/vacuum/EFT packet.

17 Evaluated survivor loci II: composite, topological, defect, solitonic, boundary, environmental, and hidden carriers

17.1 Composite and emergent carriers

Let a composite family point contain a source-coherent constituent diagram $H_{\text{const}}(\theta)$, a derivation/binding map D_θ , a derived carrier node c_θ , and a source-to-target matching map M_θ . Define

$$\mathcal{C}_{\text{comp}}(\theta) = \left\{ (D_\theta, M_\theta) : \begin{array}{l} D_\theta \text{ is source-constitutively coherent,} \\ M_\theta \text{ preserves the complete target-facing trace,} \\ \text{constituent and derived anomaly/vacuum data share one witness} \end{array} \right\}.$$

The exact composite locus is

$$\begin{aligned} \Theta_{\text{COMP}}^{\text{surv}} = \{ \theta \in \Theta_{\text{COMP}}^{\text{part}} : & \mathcal{C}_{\text{comp}}(\theta) \neq \emptyset, \mathbf{A}_{\text{COMP}}(\theta) = 0, \\ & \text{equations (34), (37) and (38) hold,} \\ & \mathcal{D}_{\text{caus}}^{\text{COMP}}(\theta) \neq \emptyset, \mathcal{V}_{\text{COMP}}(\theta) \neq \emptyset, \\ & \mathbf{R}_{\text{COMP}}(\theta) = 0, \\ & \text{Cone}_{\text{COMP}}(\theta) \neq \emptyset \}. \end{aligned} \tag{53}$$

This is an exact implicit moduli locus. It retains constituent structure whenever the carrier trace detects it; a low-energy effective field alone does not certify the common microscopic witness.

Theorem 17.1 (Composite evaluation theorem). *A source-coherent composite point survives iff it lies in $\Theta_{\text{COMP}}^{\text{surv}}$ of equation (53). A composite description whose target channel, anomaly matching, and vacuum support require different constituent theories has empty common-cone set and fails.*

17.2 Pure topological carriers

Call a topological point *pure* when its active carrier core contains topological/global state data and target attachments but no bulk dynamical process, local propagating node, hidden dilation, environment co-carrier, or non-topological law capable of generating continuous protocol dependence.

Lemma 17.2 (Pure topological response invariance). *For a pure topological point, every target-visible response obtained solely from the topological carrier is invariant under smooth baseline/energy deformations that preserve the topology and boundary incidence.*

Proof. By definition, the active physical data are topological and no metric-, energy-, or propagation-dependent source law is present. Smooth deformations preserving topology and incidence leave all source-coherent topological evaluations unchanged. Any nonconstant continuous protocol dependence would therefore require an additional dynamical law or carrier and move the point to a hybrid stratum. $\square$

Theorem 17.3 (Pure topological exclusion).

$$\Theta_{\text{TOP}}^{\text{pure, surv}} = \emptyset.$$

Proof. The frozen target contains two nonconstant boundary-visible response-mode classes. By lemma 17.2, a pure topological point cannot satisfy equation (37). Hence K_{resp} fails. A topological sector can survive only as an active co-carrier in a hybrid containing a dynamical source-to-detection channel. $\square$

Consequently,

$$\Theta_{\text{TOP}}^{\text{surv}} = \Theta_{\text{TOP} \cap \text{HYB}}^{\text{surv}}. \quad (54)$$

Topological inflow, global-form data, or defect support may remain indispensable inside this hybrid locus.

17.3 Pure extended and defect-supported carriers

Call an extended/defect point *pure* when its active core contains only a finite source-coherent defect/extended network and fixed boundary attachments, with no bulk dynamical law or carrier producing continuous source-to-detector propagation.

Theorem 17.4 (Pure defect exclusion).

$$\Theta_{\text{EXT}}^{\text{pure, surv}} = \emptyset.$$

Proof. A fixed source-coherent defect network can change sectors at its incidence loci and can carry fusion, junction, anomaly, or topological data. Without a bulk dynamical channel, however, it supplies neither the complete source-to-detection continuation required by equation (40) nor the two continuous visible response classes required by equation (37). Any defect architecture that does supply those structures contains an additional law, local/hidden carrier, environment, or extended dynamical network and is a hybrid. $\square$

Thus

$$\Theta_{\text{EXT}}^{\text{surv}} = \Theta_{\text{EXT} \cap \text{HYB}}^{\text{surv}}. \quad (55)$$

This does not make defects irrelevant: an active defect can be required for inflow, fusion, global attachment, or a target-visible channel inside the hybrid.

17.4 Solitonic and localized nonperturbative carriers

Let $\mathcal{S}_{\text{sol}}(\theta)$ be the moduli set of source-coherent finite-energy or otherwise source-admitted persistent profiles ϕ_θ with:

1. the required topological/nonlinear charge and boundary conditions;
2. a stable or source-persistent internal state space;

3. a charged-current attachment map;
4. an event-complete transport/channel construction;
5. common anomaly, vacuum, deformation, and EFT data.

The exact solitonic locus is

$$\begin{aligned} \Theta_{\text{SOL}}^{\text{surv}} = \{ \theta \in \Theta_{\text{SOL}}^{\text{part}} : \mathcal{S}_{\text{sol}}(\theta) \neq \emptyset, \mathbf{A}_{\text{SOL}}(\theta) = 0, \\ \text{equations (34), (37) and (38) hold,} \\ \mathcal{D}_{\text{caus}}^{\text{SOL}}(\theta) \neq \emptyset, \mathcal{V}_{\text{SOL}}(\theta) \neq \emptyset, \\ \mathcal{M}_{\text{SOL}}(\theta) \neq \emptyset, \mathbf{R}_{\text{SOL}}(\theta) = 0, \\ \text{Cones}_{\text{SOL}}(\theta) \neq \emptyset \}. \end{aligned} \quad (56)$$

Theorem 17.5 (Solitonic evaluation theorem). *A solitonic point survives exactly on the moduli locus equation (56). Soliton existence without target transport fails K_{chan} or K_{loc} ; a target channel without a common supporting profile/vacuum fails K_{sim} .*

17.5 Pure boundary and edge carriers

A pure boundary point has active nodes only at the source/detector or admitted edge, with no bulk continuation carrier or law connecting the separated boundary events.

Theorem 17.6 (Pure boundary exclusion).

$$\Theta_{\text{BDY}}^{\text{pure, surv}} = \emptyset.$$

Proof. The target contains distinct source and detection events joined by an observable propagation continuation. Boundary-localized data can alter preparation/effect classes or supply edge degrees of freedom, but without a bulk continuation witness they cannot satisfy equation (40). Altering the boundary semantics to remove that burden is a target update. Therefore a boundary carrier can survive only in a bulk-coupled hybrid:

$$\Theta_{\text{BDY}}^{\text{surv}} = \Theta_{\text{BDY} \cap \text{HYB}}^{\text{surv}}.$$

□

17.6 Pure environmental carriers

A pure environmental point contains an active medium/environment node but no independent system carrier or propagation law connecting the charged-current source and detector.

Theorem 17.7 (Pure environment exclusion).

$$\Theta_{\text{ENV}}^{\text{pure, surv}} = \emptyset.$$

Proof. An environment can modify or co-carry a system channel, but a pure environment with no system carrier/law supplies no event-complete active charged-current transfer map from source to detector. It therefore fails equations (34), (38) and (40). Any environmental realization that passes contains an active system law or carrier and belongs to a hybrid:

$$\Theta_{\text{ENV}}^{\text{surv}} = \Theta_{\text{ENV} \cap \text{HYB}}^{\text{surv}}.$$

□

17.7 Hidden and enlarged-state-space carriers

Let a hidden family point contain a source-coherent dilation

$$V_\theta : \mathcal{H}_{\text{act}} \longrightarrow \mathcal{H}_{\text{act}} \otimes \mathcal{H}_{\text{hid}}$$

or the corresponding general source-process object, together with projection to the visible event instrument. Define $\mathcal{H}_{\text{hid}}(\theta)$ as the set of such dilations whose hidden support, charges, anomaly data, vacuum branch, and matching maps belong to the same active carrier trace. The exact hidden locus is

$$\begin{aligned} \Theta_{\text{HID}}^{\text{surv}} = \{ \theta \in \Theta_{\text{HID}}^{\text{part}} : \mathcal{H}_{\text{hid}}(\theta) \neq \emptyset, \mathbf{A}_{\text{HID}}(\theta) = 0, \\ \text{equations (34), (37) and (38) hold,} \\ \mathcal{D}_{\text{caus}}^{\text{HID}}(\theta) \neq \emptyset, \mathcal{V}_{\text{HID}}(\theta) \neq \emptyset, \\ \mathcal{M}_{\text{HID}}(\theta) \neq \emptyset, \mathbf{R}_{\text{HID}}(\theta) = 0, \\ \text{Cone}_{\text{HID}}(\theta) \neq \emptyset \}. \end{aligned} \quad (57)$$

Invisible loss alone fails (38); an active hidden architecture can survive when its visible projection realizes the complete target response.

Theorem 17.8 (Hidden locus is nonempty).

$$\Theta_{\text{HID}}^{\text{surv}} \neq \emptyset.$$

Proof. The gauge-singlet construction used in theorem 16.5 is also an enlarged-state-space/hidden-sector realization when the singlet sector is retained explicitly and the visible active channel is obtained by projection. Its event-complete instrument includes active and null outcomes; the singlet representations are anomaly-free; and the same source architecture supplies the vacuum and matching data. The visible projection, hidden dilation, anomaly ledger, vacuum branch, and matching map satisfy the full global relation theory. Hence it defines a point of $\Theta_{\text{HID}}^{\text{surv}}$. $\square$

Table 4: Evaluated nonmatter-family results.

Stratum	Exact result	Decisive condition
Composite / emergent	Exact implicit locus equation (53).	One derivation/matching/vacuum/anomaly witness architecture.
Pure topological	Empty.	No nonconstant target-visible response without a dynamical hybrid.
Pure defect/extended	Empty.	No complete bulk continuation and visible-mode packet without a dynamical hybrid.
Solitonic	Exact implicit locus equation (56).	Persistent profile, internal response, boundary attachment, common vacuum.
Pure boundary	Empty.	No bulk source-to-detection continuation.
Pure environment	Empty.	No system event channel or active transfer.
Hidden/enlarged	Exact locus equation (57); nonempty.	Event-complete visible projection, common hidden anomaly/vacuum/matching structure.

18 Hybrid closure, actual survivor residue, and parameter-stratified survival

18.1 Hybrids as first-class candidates

PCR-3 is closed under arbitrary lawful finite hybrid formation. A hybrid may combine local, hidden, environmental, topological, defect, boundary, solitonic, composite, or law-bearing components through source-coherent active incidences. Its PCR-4 map is the complete

$$\widehat{\Phi}_{\text{HYB}}(\theta)$$

of equation (47), evaluated on the hybrid source interpretation rather than on independently selected constituent ledgers.

Theorem 18.1 (Exact hybrid survivor locus).

$$\Theta_{\text{HYB}}^{\text{surv}} = \widehat{\Phi}_{\text{HYB}}^{-1}(\mathcal{Z}_4^{\text{sem}}).$$

Proof. This is theorem 15.1 specialized to the hybrid source family. The map includes every cross-anomaly, cross-vacuum, mixed matching, activated-extension, common-cone, and global survival-relation coordinate created by the gluing. Its target is $\mathcal{Z}_4^{\text{sem}}$, the family realization of $\mathcal{A}_4^{\text{sem}}$, not a product of constituent pass loci. $\square$

Theorem 18.2 (No constituentwise survivor rule). *In general,*

$$\text{Surv}_4(X) = \text{Surv}_4(Y) = 1 \not\Rightarrow \text{Surv}_4(X \star Y) = 1,$$

and constituent failure does not automatically imply hybrid failure.

Proof. Hybrid gluing can create new anomaly terms, change the vacuum, alter the event instrument or visible response, activate generalized-symmetry constraints, or destroy a common cone. Conversely, an active co-carrier can lawfully close a constituent obstruction when the compensating incidence and witness are part of the source-coherent hybrid. Only $\widehat{\Phi}_{\text{HYB}}$ evaluates the complete architecture, including every cross-signature residual. $\square$

18.2 Pure-family reductions

The exclusion theorems of section 17 do not remove topological, defect, boundary, or environmental data from the survivor universe. They prove that those structures cannot carry the complete target alone. Their exact contributions are the hybrid loci

$$\Theta_{\text{TOP}}^{\text{surv}} = \Theta_{\text{TOP} \cap \text{HYB}}^{\text{surv}}, \quad \Theta_{\text{EXT}}^{\text{surv}} = \Theta_{\text{EXT} \cap \text{HYB}}^{\text{surv}}, \quad (58)$$

$$\Theta_{\text{BDY}}^{\text{surv}} = \Theta_{\text{BDY} \cap \text{HYB}}^{\text{surv}}, \quad \Theta_{\text{ENV}}^{\text{surv}} = \Theta_{\text{ENV} \cap \text{HYB}}^{\text{surv}}. \quad (59)$$

These intersections retain active inflow, fusion, edge, or medium data whenever the complete hybrid map requires them.

18.3 Actual evaluated survivor residue

Let FamPres_ν be the PCR-3 source-family presentation groupoid and let $\simeq_{\text{inst}}$ be complete active carrier-trace equivalence on family instances. Define

$$\mathfrak{S}_{\nu, \text{eval}}^{(4)} = (\Theta_{\text{LAW}}^{\text{surv}} \sqcup \Theta_{\text{LOCAL}}^{\text{surv}} \sqcup \Theta_{\text{COMP}}^{\text{surv}} \sqcup \Theta_{\text{SOL}}^{\text{surv}} \sqcup \Theta_{\text{HID}}^{\text{surv}} \sqcup \Theta_{\text{HYB}}^{\text{surv}}) / \simeq_{\text{inst}}. \quad (60)$$

The law, local, composite, solitonic, hidden, and hybrid terms include all family overlaps and all topological, defect, boundary, environmental, and existing-substrate contributions through the reductions already proved.

Theorem 18.3 (Evaluated-residue theorem). *The canonical survivor universe defined by the PCR-4 evaluator is exactly*

$$\mathfrak{S}_v^{(4)} = \mathfrak{S}_{v,\text{eval}}^{(4)}.$$

Proof. Every PCR-3 family point lies in one of the principal source families or a finite hybrid. The strict base stratum is empty by theorem 16.1; all other existing-substrate survivors reduce to law/composite/environment/hybrid overlaps by theorem 16.2. Pure topological, defect, boundary, and environmental strata are empty by theorems 17.3, 17.4, 17.6 and 17.7; their surviving points are already in $\Theta_{\text{HYB}}^{\text{surv}}$. The remaining principal families have the exact loci equations (51) to (53), (56) and (57) and theorem 18.1. Quotienting the disjoint presentation union by complete carrier-instance equivalence removes family overlaps without changing physical multiplicity. Hence the union is exactly $\mathfrak{S}_v^{(4)}$. $\square$

Theorem 18.4 (Proper, nonempty physical collapse). *At the declared scope,*

$$\emptyset \neq \mathfrak{S}_v^{(4)} \subseteq \mathfrak{U}_v,$$

and $\mathfrak{S}_v^{(4)}$ contains at least two carrier-inequivalent classes.

Proof. Nonemptiness follows from the law-only witness of theorem 16.3; a carrier-distinct local/hidden witness is supplied by theorems 16.5 and 17.8. Properness follows because the strict existing-substrate stratum and the pure topological, pure defect, pure boundary, and pure environment strata are nonempty PCR-3 candidate strata but have empty survivor loci. Carrier inequivalence of the two positive witnesses follows because one active trace contains no new primitive local node while the other does. $\square$

Corollary 18.5 (PCR-4 does not produce a unique survivor). *The simultaneously admissible residue is not singleton at the carrier-semantic level. Any later unique-occupancy or physical-realization result must therefore use additional role-incidence and compatibility structure rather than reinterpreting PCR-4 survival as selection.*

18.4 Exact parameter strata

For every family template u ,

$$\Theta_u^{\text{part}} = \Theta_u^{\text{surv}} \dot{\cup} \Theta_u^{\text{fail}} \dot{\cup} \Theta_u^{\text{TU}},$$

where $\Theta_u^{\text{surv}} = \widehat{\Phi}_u^{-1}(\mathcal{F}_4^{\text{sem}})$, the failure locus is the complement inside the same target fiber with its complete FailSet_u , and Θ_u^{TU} contains parameter points that change a frozen target anchor. The loci can be algebraic, semialgebraic, operator-theoretic, categorical, discrete, continuous, disconnected, or isolated.

Proposition 18.6 (No benchmark substitution). *No finite set of benchmark points proves the survival or failure of an open parameter family unless a separate theorem reduces the full source locus to those points.*

Proof. A finite sample cannot determine a general predicate on an open, disconnected, functional, or stratified domain. Benchmark calculations are witnesses or falsifiers for particular points. The full result is the preimage of the global semantic model locus under the explicit family map. $\square$

18.5 PCR-3 activity strata remain fixed

At source-null parameter points PCR-3 has already stripped inactive incidences and merged carrier-identical instances. PCR-4 evaluates that pointwise active candidate. It cannot restore a stripped sector because a later constraint would benefit from it, nor strip a source-nonnull incidence because it complicates the common witness.

Theorem 18.7 (No downstream identity mutation). *If PCR-4 selects a parameter point on which the PCR-3 active carrier trace has a particular identity, every survival and failure evaluation uses that identity. No PCR-4 result can change the trace retrospectively.*

Proof. Candidate identity is an immutable input. Source activity is a function of source semantics and the parameter point, not of the survival outcome. $\square$

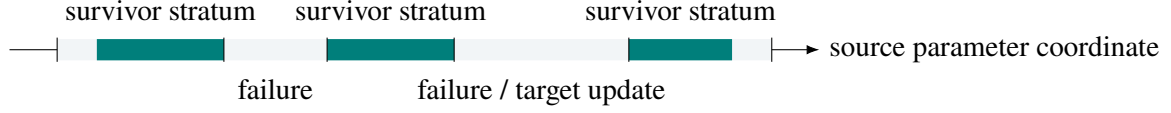


Figure 4: A symbolic PCR-3 family may have disconnected survivor strata. PCR-4 preserves the exact locus rather than replacing it by benchmark points.

Theorem 18.8 (Stratum-order theorem). *For every source family u , there is a refinement*

$$\Theta_u^{\text{part}} = \bigsqcup_a \Theta_u^{[a]} = \bigsqcup_{a,b} \Theta_u^{[a,b]},$$

where a is the PCR-3 activity type and b is the PCR-4 evaluation type. The second index may refine but cannot alter the first.

Proof. The activity partition is a source-semantic function of u and θ fixed before downstream evaluation. The PCR-4 map $\hat{\Phi}_u$ is then computed on the resulting active candidate instance. Therefore every evaluation stratum lies inside one activity stratum. $\square$

18.6 Dense and isolated survivor loci

PCR-4 does not impose genericity or naturalness. A survivor locus may be dense with empty interior, discrete, isolated, or singular. Topology of the locus alone does not confer preference. Experimental accessibility and observable separation belong to later papers unless explicitly included in the frozen target.

19 Failure classification, fixed-candidate exclusion, and omission tests

Definition 19.1 (PCR-4 failure certificate). *For $x \notin \mathfrak{S}_v^{(4)}$,*

$$\text{FailCert}_4(x) = (x, \mathcal{K}_4^{\text{act}}(x), \text{Eval}_4(x), \text{FirstFail}, \text{AllFail}, W_x, \text{ScopeDisposition}, \text{RepairClass}). \quad (61)$$

19.1 Failure classes

A nonsurviving candidate is classified by one or more of the following physical failure classes:

Failure class	Meaning
Source failure	PCR-3 source coherence, activity or candidate identity is violated; upstream reopening is required.
Role failure	The complete nested $r_\Delta \succ r_{\infty}$ burden is not occupied.
Boundary/channel failure	The charged-current boundary or event-complete active transfer is absent.
Response failure	The target visible-mode packet is not reproduced.
Transport/environment failure	No lawful source-to-detection continuation or common environmental realization exists.

Gauge/representation/ charge failure	The active carrier cannot attach consistently to the frozen substrate.
Anomaly failure	An active gauge obstruction remains or an active global anomaly is unmatched.
Vacuum/deformation failure	No common physical baseline or survivor parameter stratum exists.
EFT/matching failure	Source and target descriptions do not connect lawfully.
Extended-sector failure	An activated generalized-symmetry/defect structure is inconsistent or unattached.
Pullback failure	Componentwise successes do not form one physical realization.
Target update	Evaluation requires new target data rather than a same-scope repair.

Theorem 19.2 (Failure-certificate completeness). *Every $x \notin \mathfrak{S}_v^{(4)}$ has a nonempty, typed failure certificate, and every listed failure is supported by an active target evaluator or common-cone obstruction.*

Proof. Evaluation totality assigns every constraint a disposition. If $x \notin \mathfrak{S}_v^{(4)}$, then by definition 8.3 there is either a fail, target update, uncertified N/A or pullback failure. Constraint completeness assigns it to the taxonomy, and the evaluator supplies the witness. $\square$

19.2 Fixed-candidate failure

Theorem 8.6 makes the evaluation a property of a fixed carrier instance. A failed point may have a nearby successful point, but the two are distinct parameterized carrier instances. Likewise, a failed pure family can coexist with a successful hybrid without changing the disposition of the original candidate.

19.3 Omitted-candidate test

A proposed survivor not represented in the stated residue must provide:

1. a source-coherent active carrier architecture in the unchanged PCR-3 comparison class;
2. a PCR-3 candidate class not represented in $\mathfrak{U}_v$, or evidence that candidate completeness failed;
3. the complete PCR-2 interface and PCR-4 common witness system;
4. satisfaction of every active intrinsic and cross-signature relation;
5. an explanation of why the object is not carrier-equivalent to an existing survivor.

Otherwise it is an existing class, a failed candidate, a target update, or an upstream falsifier.

Corollary 19.3 (Negative results are scientific results). *An empty family survivor locus is a family-wide exclusion theorem at the declared scope when established by complete source and failure certificates.*

19.4 Failure localization

A complete failure description serves two scientific purposes. First, it shows that a nonsurviving candidate has not been omitted from the analysis. Second, it identifies the load-bearing theorem or physical relation responsible for exclusion. Role, anomaly, common-cone, and target-scope failures therefore have different mathematical consequences.

Definition 19.4 (First load-bearing failure). *The first load-bearing failure of x is the earliest failed predicate in the fixed dependency order whose correction would require a physical change rather than a presentation change. It is not necessarily the only failure.*

19.5 Classification of responses to failure

A proposed change following a failure falls into one of four scientific classes:

1. *exposure*: a witness or lawful equivalence already present in the same candidate was omitted from the record;
2. *redescription*: only presentation skin changes;
3. *candidate motion*: a parameter, active incidence, field content, vacuum, environment, matching map, or extended sector changes, producing another carrier instance;
4. *target/source update*: the comparison class itself changes.

There is no fifth class that changes the physical truth value while preserving every fixed scientific input.

Proposition 19.5 (Failure-report preservation). *If two carrier-equivalent instances have isomorphic witness diagrams, their complete failure sets and first load-bearing failure types agree.*

Proof. Every constraint and its dependency order are carrier-equivalence invariant. The failure set is therefore identical, and the same earliest physical obstruction is selected. $\square$

20 Survivor equivalence, interface degeneracy, and live residue

20.1 Inherited carrier equivalence

PCR-4 does not invent a new candidate identity. The survivor equivalence is inherited:

$$[x]_{\text{surv}} = [y]_{\text{surv}} \iff x \equiv_{\text{car}} y. \quad (62)$$

Theorem 20.1 (Survivor-equivalence congruence). *The survivor predicate descends to $\mathfrak{U}_v$, and the survivor subset is well defined independently of candidate representative.*

Proof. Immediate from theorem 6.4 and common-cone isomorphism invariance. $\square$

20.2 Interface equality is coarser

Two carrier-distinct survivors may satisfy

$$\Pi_v(x) = \Pi_v(y) \quad (63)$$

while $x \not\equiv_{\text{car}} y$. PCR-4 preserves them as distinct survivors because they represent different physical carrier architectures. PCR-5 may later find them equivalent or inequivalent under its lawful role-incidence action, but PCR-4 does not anticipate that result.

20.3 Presentation quotient and live residue

The PCR-4 survivor set is not yet the Role-Occupancy Closure live residue. PCR-5 must construct target-relative eligibility and joint role-assignment compatibility. A PCR-4 survivor may fail to occur in any complete compatible role assignment and therefore be non-live at PCR-5 scope.

Proposition 20.2 (Survivor multiplicity taxonomy). *PCR-4 distinguishes:*

1. *presentation multiplicity already removed by $\equiv_{\text{car}}$;*
2. *parameter multiplicity inside one carrier family;*
3. *carrier-architecture multiplicity with equal PCR-2 interfaces;*
4. *genuinely different interface behavior;*
5. *target-fiber multiplicity.*

Only the first is removed automatically at PCR-4 scope.

20.4 Automorphism preparation

For every survivor, PCR-4 exports the complete carrier trace, interface packet, witness incidence and lawful carrier automorphisms. It does not quotient by the future PCR-5 assignment automorphism group $\Gamma_{C,T}^{\text{lawful}}$.

Nonclaim 20.3 (No singleton inference). *A singleton $\mathfrak{S}_v^{(4)}$ would not by itself establish a singleton compatible role-assignment orbit. Survival and occupancy closure are different predicates.*

20.5 Three equivalence layers

The manuscript uses three distinct equivalence relations:

1. presentation equivalence, removing notation, basis, gauge representative, and other nonphysical descriptive structure;
2. carrier equivalence $\equiv_{\text{car}}$, identifying complete active carrier architectures;
3. PCR-2 interface equivalence, identifying target-facing operationally indistinguishable interfaces.

The first is contained in the second for the scientific candidate universe. The third can be strictly coarser than the second. PCR-4 survivor identity is inherited from carrier equivalence, not interface equivalence.

Theorem 20.4 (No interface-induced candidate merge). *Let $x, y \in \mathfrak{S}_v^{(4)}$ with $\Pi_v(x) = \Pi_v(y)$ but $x \not\equiv_{\text{car}} y$. Then x and y remain distinct PCR-4 survivor classes.*

Proof. The PCR-3 candidate universe is already the quotient by complete carrier-semantic identity. PCR-4 is a subobject construction on that universe and has no theorem authorizing a coarser quotient. Equal target-facing behavior does not erase different active carrier structure. A later role-assignment automorphism can identify assignment presentations only under the complete PCR-5 certificate, not at PCR-4. $\square$

20.6 Residue geometry

The survivor residue can have nontrivial automorphisms, family intersections, continuous parameter strata, and multiple carrier classes over one interface point. This geometry is retained because it can affect later eligibility and joint assignment compatibility.

21 Semantic completeness and the exact survivor theorem

21.1 Coverage and soundness maps

Let

$$q : \text{CarrierReal}_{\text{sem}}(T_v^{(3)}) \longrightarrow \mathfrak{U}_v$$

be the PCR-3 quotient map. Restricting q to semantic survivors gives

$$q_{\text{surv}} : \text{CarrierSurv}_{\text{sem}}(T_v^{(4)}) \longrightarrow \mathfrak{S}_v^{(4)}.$$

Conversely, for $x \in \mathfrak{S}_v^{(4)}$, choose any source interpretation representative X_x supplied by PCR-3 source soundness.

Theorem 21.1 (Coverage). *Every semantic survivor maps to a point of the evaluated PCR-4 residue:*

$$q(\text{CarrierSurv}_{\text{sem}}(T_v^{(4)})) \subseteq \mathfrak{S}_v^{(4)} = \mathfrak{S}_{v,\text{eval}}^{(4)}.$$

Proof. Let $X \in \text{CarrierSurv}_{\text{sem}}(T_v^{(4)})$. PCR-3 completeness supplies $x = q(X) \in \mathfrak{U}_v$ with equal active carrier trace. By semantic survival, the complete trace $\tau_4^{\text{surv}}(R_X)$ is a model of $\mathbb{S}_4^{\text{surv}}$, hence lies in $\mathcal{A}_4^{\text{sem}}$. By theorem 5.14, its intrinsic signatures, cross-signature relations, and common-cone data satisfy the

global semantic theory. Therefore the associated family parameter lies in $\widehat{\Phi}_F^{-1}(\mathcal{Z}_4^{\text{sem}})$. The reductions of theorem 18.3 place its carrier class in $\mathfrak{S}_{\nu, \text{eval}}^{(4)} = \mathfrak{S}_{\nu}^{(4)}$. $\square$

Theorem 21.2 (Soundness). *Every canonical point of the evaluated residue has a semantic survivor representative:*

$$\mathfrak{S}_{\nu}^{(4)} \subseteq q(\text{CarrierSurv}_{\text{sem}}(T_{\nu}^{(4)})).$$

Proof. Let $x = [F, \theta] \in \mathfrak{S}_{\nu}^{(4)}$. By theorem 18.3, $\theta \in \widehat{\Phi}_F^{-1}(\mathcal{Z}_4^{\text{sem}})$. PCR-3 source soundness supplies a source-coherent, currently active semantic carrier representative X_x . Membership in $\mathcal{Z}_4^{\text{sem}}$ gives an event-complete instrument, the complete target record and mode packet, lawful boundary attachment, causal/environment support, zero anomaly residual, a common vacuum branch, a matching diagram, every activated extension condition, a nonempty physical common-cone set, and $\mathbf{R}_F(\theta) = 0$. These are exactly the conditions of definition 4.1. Hence $X_x \in \text{CarrierSurv}_{\text{sem}}(T_{\nu}^{(4)})$ and $q(X_x) = x$. $\square$

Theorem 21.3 (Exact PCR-4 survivor universe). *The quotient map induces a canonical bijection*

$$\boxed{\text{CarrierSurv}_{\text{sem}}(T_{\nu}^{(4)}) / \equiv_{\text{car}} \cong \mathfrak{S}_{\nu}^{(4)} = \mathfrak{S}_{\nu, \text{eval}}^{(4)}} \quad (64)$$

Proof. Surjectivity follows from theorem 21.2; coverage follows from theorem 21.1. If two semantic survivors map to the same $x \in \mathfrak{U}_{\nu}$, PCR-3 quotient exactness gives carrier equivalence, proving injectivity. Evaluation congruence and theorem 5.13 make the map independent of representatives and of any finite tabulation. $\square$

Corollary 21.4 (Pointwise family form).

$$\mathfrak{S}_{\nu}^{(4)} \cong \left(\bigsqcup_{u \in \text{FamPres}_{\nu}} \widehat{\Phi}_u^{-1}(\mathcal{Z}_4^{\text{sem}}) \right) / \simeq_{\text{inst}},$$

with the empty pure strata and family-overlap reductions of sections 16 to 18 imposed before quotienting.

Corollary 21.5 (Nontrivial evaluated collapse).

$$\boxed{\emptyset \neq \mathfrak{S}_{\nu}^{(4)} \subsetneq \mathfrak{U}_{\nu}}$$

and $\mathfrak{S}_{\nu}^{(4)}$ contains at least two carrier-inequivalent survivor classes.

PCR-4 endpoint

PCR-4 has performed simultaneous survival at the declared scope. The independent survival-consistency theory is specialized to every PCR-3 source family through the global maps $\widehat{\Phi}_F$. The strict existing-substrate, pure topological, pure defect, pure boundary, and pure environment strata are eliminated; law-only and local/hidden survivor points are explicitly witnessed; composite, solitonic, and arbitrary hybrid survivors are fixed by exact source-level moduli and cross-relation equations. Every semantic survivor is represented in the canonical evaluated residue, every canonical survivor has a semantic physical representative, and every non-survivor has a nonempty typed failure set. No role assignment or physical realization has been inferred.

21.2 Why the theorem is not definitional

Three independent constructions meet in theorem 21.3:

1. the semantic survivor class is defined before any family enumeration or disposition;
2. the global relation theory $\mathbb{S}_4^{\text{surv}}$ is derived before the finite PCR-4 normal form;

3. the canonical residue is computed as the quotient of the explicit global family preimage loci $\widehat{\Phi}_F^{-1}(\mathcal{Z}_4^{\text{sem}})$. The theorem therefore does more than identify two notations for the same predicate. It proves that the source-first carrier universe undergoes a proper, nonempty physical collapse and gives the exact surviving strata.

21.3 Falsifiers

The capstone is reopened by any of the following:

1. a source-coherent same-target carrier omitted from $\mathfrak{U}_\nu$;
2. a same-target constitutive or bridge relation on existing trace sorts not generated by $\mathbb{S}_4^{\text{surv}}$;
3. an intrinsic signature or cross-signature relation not represented by $\widehat{\Phi}_F$;
4. a carrier-equivalent pair receiving different global family-evaluation signatures;
5. a canonical survivor lacking a semantic physical representative;
6. a semantic survivor lying outside every explicit global family locus;
7. a witness-split tuple incorrectly admitted as a common-cone realization;
8. a marginally admissible common-cone packet with $\mathbf{R}_F \neq 0$ incorrectly admitted;
9. a claimed empty pure stratum with an actual target-visible dynamical channel satisfying $\mathcal{Z}_4^{\text{sem}}$.

A concrete falsifier triggers revision; it is not relabelled away.

Nonclaim 21.6 (No Role-Occupancy Closure). *The capstone is an exact survivor theorem. It does not prove*

$$|\Sigma_C(T)/\Gamma_{C,T}^{\text{lawful}}| = 1.$$

PCR-5 must independently construct target-relative eligibility, joint assignment compatibility, live residue, and lawful role-incidence automorphisms.

21.4 Independent completeness burdens

The capstone composes:

1. PCR-1 completeness of the forced role burden at the target scope;
2. PCR-2 completeness of the target-facing evaluator/interface grammar;
3. PCR-3 completeness of the source-first carrier universe;
4. PCR-4 completeness of the independent survival-consistency theory;
5. PCR-4 exact specialization of its intrinsic and cross-signature relations to every source family and hybrid stratum.

The final isomorphism fails if any layer is replaced by a finite model list, expected outcome, product of marginal pass sets, or stored survivor label.

21.5 No hidden survivor multiplicity

Every semantic survivor multiplicity is represented by the canonical carrier quotient, and every canonical multiplicity corresponds to carrier-semantic multiplicity. Presentation copies and source-family overlaps do not inflate the residue. Conversely, interface-degenerate but carrier-distinct law-only and local/hidden points remain distinct survivors for PCR-5.

22 Robustness tests and scope boundaries

The survivor theorem is most informative when its logical layers are separated by explicit boundary cases. The cases below are chosen to distinguish channel adequacy from carrier identity, marginal consistency from simultaneous realization, and same-target evaluation from a change of target or source theory.

Table 6: Representative robustness tests.

Case	Mathematical disposition
Active transfer with invisible loss	Loss is included in the event instrument but does not substitute for charged-current transfer.
Response modes confined to an unattached sector	Fails the role/common-cone requirement.
Channel and anomaly calculations using different active representations	Marginal calculations may pass, but the shared-witness pullback fails.
Anomaly cancellation by a spectator sector	Valid only if the sector is physically active in the PCR-3 carrier trace.
Weak but nonzero hidden coupling	The sector remains active; no numerical threshold defines carrier participation.
Source-null coupling at a special parameter point	Candidate identity is already reduced by PCR-3 activity semantics before survival is evaluated.
Integrated-out local carrier versus a law-only effective description	The target interface may coincide while carrier-semantic identity remains distinct.
Defect or topological anomaly inflow	Contributes only when the corresponding extended sector is source-coherent and physically attached.
Detector-localized change of preparation or measurement semantics	Defines a target update unless it preserves the fixed operational boundary.
Pairwise-compatible but globally inconsistent witness packet	Lies outside the shared-witness pullback.
Composite candidate with an acceptable reduced response but no coherent supporting carrier	Fails source, deformation, or common-witness consistency.
Environmental realization using incompatible medium states	Fails the common environmental/vacuum realization condition.
Family checked only at sampled benchmark points	Does not establish a family theorem; the exact parameter locus remains open.
Small but nonzero anomaly residual	Fails unless the fixed target itself contains a physical tolerance.
Point-local candidate insensitive to global-form data	The corresponding extended-sector condition is inert.
Malformed non-invertible defect associator	Excluded already by source-constitutive coherence.

Case	Mathematical disposition
Physical modification after a failed evaluation	Defines another carrier instance or target rather than changing the result for the same candidate.
Singleton survivor residue	Would still not imply licensed occupancy or physical realization.

22.1 Marginal consistency versus a global physical realization

A central negative example is a packet whose propagation, anomaly, vacuum, and matching data are each individually admissible but cannot be obtained as restrictions of one physical architecture. This demonstrates that the raw product of marginal solution spaces is structurally too large. The common-cone construction is therefore not an optional bookkeeping device: it is the mathematical condition that the successful component statements refer to one and the same candidate.

Even common-cone existence is not by itself sufficient. Section 5 shows that cross-signature relations can fail within a single co-instantiated architecture. Global consistency therefore requires both physical co-instantiation and satisfaction of the full relation theory.

22.2 Operational scope of the propagation channel

The event-complete instrument is a target-operational object. It does not assert that every microscopic theory is fundamentally a linear CPTP evolution on a three-dimensional neutrino Hilbert space. Unitary dynamics, open-system reductions, dilations, decay channels, environmental couplings, and other admitted source constructions may induce the same target-level event instrument. A genuinely richer operational process category would define an explicit extension of the comparison class.

22.3 Scope of source completeness

The carrier-universe theorem used here is relative to the finite occupancy-bearing presentation regime established upstream. A genuinely essentially infinitary carrier architecture that does not admit such a presentation lies outside that source-theory scope and would require a separate extension. This limitation is part of the theorem statement rather than an exclusion by enumeration.

22.4 Use of empirical information

The target-visible response packet is imported from the previously fixed experimental analysis and is not refitted separately for each candidate. The present paper asks whether a candidate realizes that fixed operational target under the admitted equivalences. A revision of the empirical target would reopen the upstream target definition; comparative model preference is not a survival criterion here.

22.5 Further diagnostic constructions

Same reduced channel, different anomaly sectors. Two carrier architectures can induce the same operational channel while differing in hidden or chiral content. They remain carrier-distinct, and only those satisfying the complete anomaly and overlap relations survive.

Vacuum branch crossing. A family may satisfy role/channel conditions on one vacuum branch and anomaly/effective-theory conditions on another. The survivor locus is the intersection on a common branch, not the union of branchwise successes.

Source-null topological coupling. At a source-null parameter point, an inactive topological co-carrier is removed by the upstream activity reduction. At a nonnull point the resulting hybrid is a distinct carrier instance and is evaluated with its full inflow, fusion, vacuum, and matching structure.

Boundary modification. A candidate may reproduce an observational record by changing detector edge modes or preparation classes. If those changes alter the fixed boundary semantics, they define a different target even when the resulting phenomenology is similar.

Isolated exact survivor. An anomaly or matching system may possess an isolated exact solution that a numerical scan misses. The parameter-locus theorem therefore requires exact symbolic characterization or certified coverage whenever sampling alone is not complete.

Interface-degenerate carrier pair. Two carrier-distinct architectures may project to the same target-facing interface and satisfy every survival relation. Both remain in the survivor residue because interface equivalence is intentionally coarser than carrier equivalence.

22.6 Direct falsification criteria

The capstone correspondence would be falsified by either of two same-target constructions: a semantic survivor with one complete physical witness architecture that belongs to the frozen carrier universe but is absent from the canonical residue, or a canonical survivor whose complete witness packet has no semantic physical representative. A same-target survival relation on the fixed trace sorts that is absent from the asserted relation theory would likewise falsify evaluator completeness. These are direct scientific counterexamples to coverage, soundness, or relation exhaustion rather than changes of target or source ontology.

23 Conclusion and consequences for role-occupancy analysis

The analysis begins from a previously fixed role complex, a complete target-relative interface, and a source-first carrier universe. The present paper adds the distinct requirement of simultaneous physical consistency. The central methodological point is that evaluator completeness and family evaluation are separate burdens: the many-sorted survival theory is derived independently of the family outcomes, and only then specialized to each carrier family through explicit intrinsic, relational, and common-witness maps.

The resulting survivor residue is

$$\mathfrak{S}_\nu^{(4)} = \mathfrak{S}_{\nu, \text{eval}}^{(4)} = \left(\bigsqcup_u \widehat{\Phi}_u^{-1}(\mathcal{X}_4^{\text{sem}}) \right) / \simeq_{\text{inst}},$$

with the reductions among overlapping source families established in the preceding sections. At the declared scope,

$$\emptyset \neq \mathfrak{S}_\nu^{(4)} \subsetneq \mathfrak{U}_\nu$$

and

$$\text{CarrierSurv}_{\text{sem}}(T_\nu^{(4)}) / \equiv_{\text{car}} \cong \mathfrak{S}_\nu^{(4)}.$$

The undeformed massless-neutrino Standard Model, pure topological, pure defect, pure boundary, and pure environmental strata are excluded as stand-alone realizations. Topological, defect, boundary, and environmental structures can nevertheless remain active inside complete dynamical hybrids. A gauge-invariant law-only completion and active local/hidden singlet completions provide explicit carrier-distinct survivor witnesses, while composite, solitonic, and general hybrid families are retained through their exact source-level moduli loci.

23.1 Physical interpretation of the residue

The residue is not singleton at the carrier-semantic level. At least a law-only completion on the existing field substrate and an active local/hidden singlet completion survive as distinct carrier architectures. The simultaneous-consistency theorem therefore neither predicts new matter nor selects a unique neutrino mechanism.

What it does establish is sharper:

1. the undeformed massless-neutrino Standard Model is not a survivor of the fixed target;
2. several broad pure non-dynamical carrier strata cannot realize the target on their own;
3. simultaneous consistency reduces the complete source-first carrier universe to a proper nonempty subset;
4. the remaining multiplicity is physical carrier multiplicity rather than a difference of notation, basis, or target-facing presentation.

23.2 Consequences for compatible role occupancy

The next mathematical question is not further survival filtering but whether the surviving carrier architectures can occupy the ordered role complex in a complete compatible assignment. That requires independently defined eligibility, joint-assignment compatibility, live-residue, and lawful assignment-automorphism data,

$$E_C, \quad J_C, \quad \Sigma_C(T), \quad \Gamma_{C,T}^{\text{lawful}},$$

and ultimately the orbit count

$$\left| \Sigma_C(T) / \Gamma_{C,T}^{\text{lawful}} \right|.$$

The present theorem does not prejudice that value. Some survivors may fail to occur in any complete compatible assignment; multiple carrier-distinct survivors may occupy the same lawful assignment orbit; or genuine underdetermination may remain.

Theorem boundary

The paper proves exact simultaneous survivor closure relative to the fixed role complex, interface, carrier universe, target, source ontology, and comparison class. It does not establish a particle prediction or physical realization. Such a conclusion would require a subsequent role-occupancy theorem and an observable-projection argument that remove every remaining lawful alternative.

23.3 Scope variation

The residue obtained here is nonempty. For a different target or source regime, an empty residue would identify an inconsistency or incompleteness among the fixed role packet, interface grammar, carrier ontology, survival relation theory, or target assumptions. The typed failure classification developed in Section 19 identifies which premise must be reconsidered.

23.4 Independence from later assignment structure

Theorem 23.1 (No assignment-induced survivor mutation). *Let A be any later role-assignment object constructed from the fixed survivor residue. No property of A can change membership in $\mathfrak{S}_v^{(4)}$ without changing the fixed inputs of the present theorem.*

Proof. $\mathfrak{S}_v^{(4)}$ is fixed by the explicit family evaluation maps and the exact semantic/canonical survivor theorem before A exists. Assignment properties are functions of the frozen residue and role-incidence data. Feeding them back into survivor membership is circular and violates the import lock. $\square$

The theorem above fixes the direction of dependence: later assignment structure may be evaluated on the survivor residue, but it cannot be used retroactively to change which carrier instances survive the present simultaneous-consistency problem.

23.5 Final statement

Within the stated target and comparison class, the simultaneous physical filtering problem is closed by the global survival-consistency theory, the explicit family evaluation maps, and the exact semantic/canonical

survivor correspondence. The remaining scientific question is whether the non-singleton survivor residue admits exactly one lawful compatible role-assignment orbit, several such orbits, or none. That question is logically downstream of the result proved here.

A Constraint normal forms and failure conditions

The named constraint families provide a compact normal-form presentation of the independently derived survival-consistency theory $\mathbb{S}_4^{\text{surv}}$. Each family specifies its physical domain, active witnesses, pass condition, failure condition, and overlap requirements. The catalogue below is therefore a summary of the theorem-level conditions rather than an independent source of scientific authority.

Table 7: Constraint normal forms for simultaneous survival.

ID	Burden	Success condition	Failure condition
K_SRC	source identity/coherence.	PCR-3 source coherence, activity and parameter identity are preserved.	Malformed or incorrectly reduced source architecture; upstream PCR-3 reopening.
K_TARGET	target fidelity.	All frozen target anchors and the charged-current boundary are preserved.	Changed boundary, record semantics, operator category or comparison class.
K_ROLE	nested role capability.	One architecture realizes the complete nested role burden.	Transfer-only, one-mode-only, or separated role realizations.
K_CC	charged-current boundary.	Preparation and detection semantics coincide with the frozen boundary.	Source/detection retargeting or detector-only reinterpretation.
K_CHAN	event-complete active transfer.	A complete physical event instrument exists and active transfer is distinguished from null loss.	Incomplete outcomes, loss-only response, or unsupported map.
K_RESP	visible response multiplicity.	The full target visible-response packet is reproduced under lawful transports.	Missing target mode class or candidate-specific recalibration.
K_LOC	locality/transport.	A lawful source-to-detection support and continuation structure exists.	Pathless influence, missing support map, or undeclared nonlocal extension.
K_ENV	environment.	Every active environmental contribution is source-defined and jointly realized.	Environment used as selector or incompatible environmental states.
K_GAUGE	gauge attachment.	The candidate attaches to the known gauge substrate without silent retargeting.	Unattached or silently enlarged gauge architecture.
K_REP	representation identity.	The same active representation is used across propagation and consistency calculations.	Representation split across calculations.
K_CHARGE	charge witness.	Charge witnesses and normalization classes are physically consistent.	Incompatible charge witnesses or illicit normalization selector.
K_ANOM	anomaly consistency.	Every active gauge obstruction is canceled and every active global anomaly is matched.	Uncanceled active obstruction or unattached inflow/spectator laundering.
K_VAC	vacuum support.	One non-null physical baseline supports all active structures.	Incompatible baseline states or hidden vacuum repair.
K_DEF	deformation/stratum.	The claimed point lies on an exact deformation/survivor stratum.	No exact stable locus or post-hoc parameter repair.

ID	Burden	Success condition	Failure condition
K_EFT	matching/decoupling.	A lawful source-to-target matching map exists.	No matching, scale retargeting, or carrier creation by matching.
K_EXT	activated generalized sector.	Every activated extended sector is coherent, relevant and attached.	Inconsistent or target-inert extended data used as repair.
K_REL	global intrinsic/cross-signature relations.	The complete survival trace is a model of every source-interface, channel-record, continuation-channel, gauge-boundary, representation-charge-anomaly, vacuum-deformation-EFT, extension-attachment, and related lawful equation.	Marginally admissible data or a common cone exists, but one frozen same-target relation fails.
K_SIM	shared witness cone.	One physical common cone realizes the complete packet.	Witness-split packet or noncommutative overlap maps.
K_NO_REPAIR	irreversibility.	No failed datum is altered without changing candidate or target identity.	Same candidate retroactively changed after failure.
K_EQUIV	carrier-equivalence congruence.	Dispositions descend to PCR-3 carrier-equivalence classes.	Carrier-equivalent representatives receive different outcomes.

A.1 Inapplicability

A constraint may be marked N/A only if its physical input sort is absent from the frozen target and candidate and the absence is itself certified. For example, a higher-form anomaly test can be N/A in a point-local candidate with no admitted line/surface sector; it cannot be N/A after the candidate activates such a sector.

A.2 Target changes

A target-update disposition is mandatory when evaluation requires new source/detection semantics, a new primitive target-facing datum, a new operator category, or a different background/fiber. Target update is neither pass nor fail inside the old target.

A.3 Independence from model preference

The success conditions depend only on source and physical witnesses together with the fixed evaluator. Model names, expected outcomes, and phenomenological preference are not inputs to the survivor predicate. This independence follows from the relation-theory construction proved in the main text.

B Family-level survivor loci

This appendix collects the family-level results derived from the universal map $\widehat{\Phi}_F$ and the exact loci established in sections 16 to 18. The table summarizes those theorems without replacing their proofs.

Table 8: Family-level structure of the survivor residue.

Family	Principal result	Exact survivor locus	Supporting result
BASE	Strict base empty; survivors only in overlaps.	$\Theta_{\text{BASE}}^{\text{surv}} = (\Theta_{\text{BASE} \cap \text{LAW}}^{\text{surv}} \cup \Theta_{\text{BASE} \cap \text{COMP}}^{\text{surv}} \cup \Theta_{\text{BASE} \cap \text{ENV}}^{\text{surv}} \cup \Theta_{\text{BASE} \cap \text{HYB}}^{\text{surv}}) / \sim$.	Theorems 16.1 and 16.2.
LAW	Exact nonempty locus.	Equation (51).	Gauge-invariant dimension-five operator witness, theorem 16.3.

Family	Principal result	Exact survivor locus	Supporting result
LOCAL	Exact nonempty locus.	Equation (52).	Gauge-singlet local-fermion witness, theorem 16.5.
COMP	Exact implicit moduli locus.	Equation (53).	One shared constituent, derivation, matching, anomaly, vacuum, and common-cone architecture.
TOP	Pure stratum empty; hybrid only.	$\Theta_{\text{TOP}}^{\text{surv}} = \Theta_{\text{TOP} \cap \text{HYB}}^{\text{surv}}$.	Lemma 17.2 and theorem 17.3.
EXT	Pure stratum empty; hybrid only.	$\Theta_{\text{EXT}}^{\text{surv}} = \Theta_{\text{EXT} \cap \text{HYB}}^{\text{surv}}$.	Theorem 17.4.
SOL	Exact implicit moduli locus.	Equation (56).	Persistent profile, internal response, target attachment, common vacuum.
BDY	Pure stratum empty; hybrid only.	$\Theta_{\text{BDY}}^{\text{surv}} = \Theta_{\text{BDY} \cap \text{HYB}}^{\text{surv}}$.	Theorem 17.6.
ENV	Pure stratum empty; hybrid only.	$\Theta_{\text{ENV}}^{\text{surv}} = \Theta_{\text{ENV} \cap \text{HYB}}^{\text{surv}}$.	Theorem 17.7.
HID	Exact nonempty locus.	Equation (57).	Active singlet-sector dilation witness, theorem 17.8.
HYB	Exact universal hybrid locus.	$\Theta_{\text{HYB}}^{\text{surv}} = \widehat{\Phi}_{\text{HYB}}^{-1}(\mathcal{Z}_4^{\text{sem}})$.	Theorem 18.1.

B.1 Universal signature matrix

For every family point, the evaluation map contains the following source-derived signatures:

$$(\Phi_{\text{CC}}, \Phi_{\text{inst}}, \Phi_{\text{rec}}, \Phi_{\text{mode}}, \Phi_{\text{tr}}, \mathbf{A}, \mathcal{V}, \mathcal{M}, \mathcal{X}, \text{Cone}).$$

The corresponding pass conditions are:

1. gauge-compatible charged-current attachment;
2. complete positivity and event completeness;
3. equality of every frozen record evaluation and lawful inclusion of the target mode packet;
4. nonzero active charged-current transfer;
5. source-to-detection causal/support continuation;
6. vanishing complete anomaly residual on one representation/charge ledger;
7. a nonempty common vacuum/deformation moduli set;
8. a nonempty source-to-target matching set;
9. complete consistency of every activated extension sector;
10. a nonempty physical common-cone set.

B.2 Actual residue

The evaluated residue is

$$\mathfrak{S}_{\nu, \text{eval}}^{(4)} = (\Theta_{\text{LAW}}^{\text{surv}} \sqcup \Theta_{\text{LOCAL}}^{\text{surv}} \sqcup \Theta_{\text{COMP}}^{\text{surv}} \sqcup \Theta_{\text{SOL}}^{\text{surv}} \sqcup \Theta_{\text{HID}}^{\text{surv}} \sqcup \Theta_{\text{HYB}}^{\text{surv}}) / \simeq_{\text{inst}}.$$

It is proper and nonempty by theorem 18.4.

C Provenance of imported results

Source	Role in the present theorem	Scope boundary
PCR-1	Ordered role complex and empirical target evidence.	No carrier interpretation or mechanism.
PCR-2	Complete interface, target trace, evaluator grammar, common-incidence profile.	No candidate generation or realization.
PCR-3	Complete carrier universe, source coherence, activity and carrier equivalence.	No interface pass/fail or survival.
Kernel/AMetric	Fixed-domain, no-repair, no-selector and standing conservation.	No neutrino-specific physics.
RPI	Locality, transport, fields, particles, vacuum, time and causality roles.	No candidate-specific dynamics.
SR1	Shared-witness pullback and no free factorization.	Structural precedent rather than a neutrino survivor result.
SYM-9/SYM-11	Global-form/generalized-symmetry scope and target typing.	Relevance-gated, not universally imposed.
ROC	Later eligibility/compatibility/assignment theorem.	Not used to promote survivors to occupants.
Kraus–Stinespring–Lindblad	Ordinary open-system/event-channel mathematics.	Target-operational representation only.
Anomaly, EFT and generalized-symmetry sources	Standard physical consistency languages.	Applied only when corresponding target/candidate sector is active.
Weinberg dimension-five operator	Explicit nonempty law-only survivor witness.	Existence witness only; no unique mechanism or selection.
Minkowski singlet construction	Explicit nonempty local/hidden survivor witness.	Existence witness only; no unique mechanism or selection.

D Supplementary proofs for simultaneous constraint closure

D.1 Evaluation totality as a sum-type theorem

For each intrinsic or cross-signature relation family let D_j be its frozen input domain and $A_j(x)$ the proposition that the candidate activates that domain. The total evaluator is

$$\widehat{E}_j(x) = \begin{cases} \text{PASS,} & A_j(x) \text{ and the relation holds,} \\ \text{FAIL,} & A_j(x) \text{ and the relation fails,} \\ \text{N/A,} & \neg A_j(x) \text{ and absence is source-certified,} \\ \text{TARGET UPDATE,} & \text{evaluation requires a new primitive target/source datum.} \end{cases}$$

Fixed target typing makes these branches disjoint. The independent theory $\mathbb{S}_4^{\text{surv}}$ and theorem 5.11 make them exhaustive. This is the mathematical content of theorem 6.3; no implementation convention enters the theorem.

D.2 Constraint completeness from the independent relation theory

The many-sorted theory $\mathbb{S}_4^{\text{surv}} = (\mathfrak{D}_4, \mathfrak{M}_4, \mathfrak{E}_4)$ is fixed before the named constraint bundle. Its models satisfy both intrinsic admissibility and the cross-signature equations R1–R10 of definition 5.6. The finite normal form contains intrinsic generators, the relation packet K_{rel} , and the one-completion generator K_{sim} . By theorem 5.13, their lawful consequence closure equals the semantic survival-evaluator class. By theorem 5.14, modelhood in $\mathbb{S}_4^{\text{surv}}$ is equivalent to passing every semantic evaluator. This proves theorem 8.4 and closes both finite-checklist and product-of-marginals circularity.

D.3 Why common-cone existence is not relation completeness

A common apex establishes that component witnesses are projections of one physical object. It does not force every target/source equation among their values. The mutation in proposition 5.15 and theorem 8.5 keeps admitted component objects and a common apex but breaks EFT/channel-record naturality. The packet is rejected by K_{rel} . This separates co-instantiation from global consistency.

D.4 Specialization to family parameter spaces

For each PCR-3 family F , the intrinsic map Φ_F collects the source-specialized instrument, record, mode, transfer, attachment, anomaly, continuation, vacuum, matching, extension, and common-cone signatures. The relation packet $\mathbf{R}_F$ evaluates the family specialization of $\mathfrak{E}_4$. The global map is

$$\widehat{\Phi}_F = (\Phi_F, \mathbf{R}_F),$$

and the semantic model locus is $\mathcal{Z}_4^{\text{sem}}$. Consequently

$$\Theta_F^{\text{surv}} = \widehat{\Phi}_F^{-1}(\mathcal{Z}_4^{\text{sem}})$$

by theorem 15.1. This preimage is an evaluated locus because every intrinsic coordinate and every cross relation is given by the physical relations in section 15; it is not the tautological comprehension “all points that survive.”

D.5 Positive-witness relation closure

For the law-only Weinberg-operator witness, the source operator, electroweak vacuum, propagation Hamiltonian, charged-current projectors, event channel, record/mode evaluation, anomaly ledger, and identity EFT matching are restrictions of one low-energy construction. Thus $\mathbf{R}_{\text{LAW}} = 0$. For the local singlet witness, the Yukawa/Majorana source theory, active-hidden channel, anomaly-free singlet representation, common vacuum, and retained-or-integrated-out matching diagram commute, so $\mathbf{R}_{\text{LOCAL}} = 0$. The hidden witness inherits the same relation closure. Hence the positive examples lie in the global model locus rather than only in its marginal projections.

D.6 Common-cone invariance

Suppose $x \equiv_{\text{car}} y$. Carrier equivalence gives an isomorphism of active source diagrams. PCR-2 semantic interface equivalence gives an isomorphism of interface witness diagrams. Their product preserves every overlap map used by W_{sim} , while source/target naturality preserves $\mathbf{R}_F$. Hence common-cone existence, relation satisfaction, Pull_4 , and Surv_4 are invariant.

D.7 Pairwise insufficiency in diagram form

Let D_P, D_A, D_V be three witness diagrams and let pairwise cones exist over each pair. A global cone additionally requires the induced maps on triple overlaps to agree. Choose pairwise maps whose composite

around the triangle is a nontrivial automorphism. Then every pair is compatible while the triple cocycle fails. Even when a global cone exists, a separate cross relation can fail, as in [proposition 5.15](#). Thus pairwise compatibility, common-cone existence, and global consistency are three distinct burdens.

D.8 Fail-closedness and candidate identity

If a candidate point $[u, \theta]$ fails and a proposed modification changes θ , source activity, incidence, representation, environment, global form, target attachment, or a cross-relation value, the modified object is another PCR-3 point or a new target. If none changes, all semantic evaluators are unchanged and the failure persists. Therefore there is no third same-instance route that changes the physical disposition while preserving the fixed candidate and target.

D.9 Parameter-locus exactness

The survivor locus is the inverse image of the global semantic model subobject under the complete family evaluation map:

$$\Theta_u^{\text{surv}} = \widehat{\Phi}_u^{-1}(\mathcal{Z}_4^{\text{sem}}).$$

No topological or measure-theoretic regularity is assumed. The locus may be algebraic, semialgebraic, functional-analytic, categorical, disconnected, or otherwise stratified within the source-defined parameter category. Reparameterization invariance follows from evaluation congruence.

D.10 Properness and nonemptiness

Properness follows from the strict-base and pure-family exclusions. Nonemptiness follows independently from the law-only, local, and hidden positive witnesses after their relation packets are shown to vanish. Carrier-trace inequivalence between the law-only and local/hidden witnesses gives at least two carrier-distinct survivor classes. Hence the evaluated collapse is both proper and nonempty rather than a formal possibility statement.

E Family-specialized evaluator maps

This appendix collects the source-family specializations of the universal map $\widehat{\Phi}_F$ from [section 15](#). The map is evaluated on the complete source-coherent parameter locus of each carrier family, and its survivor locus is the exact preimage $\widehat{\Phi}_F^{-1}(\mathcal{Z}_4^{\text{sem}})$. The results are defined on the full symbolic domains rather than by finite benchmark sampling.

E.1 Existing-substrate family

A base-family point has no new primitive carrier node. Its complete evaluation is the restriction of $\widehat{\Phi}_{\text{BASE}}$ to the existing-substrate source domain. The strict undeformed point has central active-neutrino propagation and therefore fails K_{role} , K_{chan} , and K_{resp} by [theorem 16.1](#). Every surviving base-labelled point is already represented by a law-only, composite, environmental, or hybrid source architecture, giving the reduction of [theorem 16.2](#).

E.2 Law-only family

For

$$X_{\text{law}}(\theta) = (\mathfrak{B}_{\text{fixed}}, \mathcal{L}_\theta, \text{Support}_\theta, \text{Attach}_{\text{CC}}),$$

the exact survivor locus is [equation \(51\)](#). It requires an event-complete instrument, the frozen record and mode signatures, active transfer, causal support, a common vacuum/deformation witness, lawful EFT matching, a vanishing cross-relation packet, and a physical common cone. [Theorem 16.3](#) supplies a nonempty source-coherent point through the gauge-invariant dimension-five lepton–Higgs operator.

E.3 Local-carrier family

A local point contains a local carrier Φ , one representation/charge datum, source couplings, charged-current attachment, a common vacuum, and matching data. Its exact locus is equation (52); the complete anomaly vector equation (41) is evaluated on the same active carrier content used by the propagation channel. Theorem 16.5 gives a source-coherent nonempty point using at least two gauge-singlet right-handed fermions with Yukawa attachment and optional Majorana data.

E.4 Composite and emergent family

A composite point contains constituents, a source-coherent derivation or binding object, a persistent derived carrier, and target attachment. The exact locus equation (53) requires nonempty derivation moduli, constituent/effective anomaly agreement, a complete event instrument, target records and modes, causal support, one vacuum, and one common physical cone. Nonemptiness is not asserted independently of those moduli equations.

E.5 Topological and defect families

A pure topological point has no target-visible dynamical channel varying over the propagation comparison variable and is eliminated by theorem 17.3. A pure defect/extended point with no dynamical bulk continuation is eliminated by theorem 17.4. Their admissible residues are therefore the hybrid reductions equations (54) and (55), where topological or defect data may remain active co-carriers inside a complete dynamical architecture.

E.6 Solitonic family

A solitonic point contains supporting fields, a persistent localized profile, internal collective coordinates, target transport, and source-defined vacuum/matching data. Its exact implicit moduli locus is equation (56). A static profile alone is insufficient; the same solitonic architecture must support the event channel, response packet, anomaly ledger, continuation, vacuum, EFT matching, and common cone.

E.7 Boundary and environmental families

A pure boundary point cannot replace the bulk source-to-detection continuation and is eliminated by theorem 17.6. A pure environment point lacks the system event channel and active flavor-transfer bearer and is eliminated by theorem 17.7. Both may survive as active co-carriers in a hybrid when the full universal map passes.

E.8 Hidden/enlarged-state-space family

A hidden point contains an enlarged physical space, source-coherent dilation or embedding, observable projection, complete event instrument, anomaly/vacuum data, and matching. Its exact locus is equation (57); theorem 17.8 gives a nonempty gauge-singlet enlarged-state-space witness. Loss-only points remain outside the locus because null outcomes do not supply r_{pct} .

E.9 Hybrid family

A hybrid point is a finite source-coherent colimit of constituent carrier diagrams with explicit gluing maps. Its exact survivor locus is

$$\Theta_{\text{HYB}}^{\text{surv}} = \widehat{\Phi}_{\text{HYB}}^{-1}(\mathcal{Z}_4^{\text{sem}})$$

by theorem 18.1. The universal global map includes every constituent burden plus the complete relation packet, including cross-anomaly, cross-vacuum, cross-matching, extension, response, and common-cone conditions created by the gluing. Passing constituents neither guarantee nor are required individually for a passing hybrid; only the complete colimit object is evaluated.

References

- [1] O. Aharony, N. Seiberg, and Y. Tachikawa. “Reading Between the Lines of Four-Dimensional Gauge Theories”. In: *Journal of High Energy Physics* 2013.8 (2013), p. 115. arXiv: [1305.0318](#).
- [2] A. Bilal. “Lectures on Anomalies”. In: (2008). arXiv: [0802.0634 \[hep-th\]](#).
- [3] H.-P. Breuer and F. Petruccione. *The Theory of Open Quantum Systems*. Oxford University Press, 2002.
- [4] Y. Choi, H. T. Lam, and S.-H. Shao. “Non-Invertible Global Symmetries in the Standard Model”. In: *Physical Review Letters* 129 (2022), p. 161601. arXiv: [2205.05086](#).
- [5] C. Cordova and S. Koren. “Higher Flavor Symmetries in the Standard Model”. In: *Annalen der Physik* 535 (2023), p. 2300031. arXiv: [2212.13193](#).
- [6] Daya Bay Collaboration, F. P. An, et al. “Spectral Measurement of Electron Antineutrino Oscillation Amplitude and Frequency at Daya Bay”. In: *Physical Review Letters* 112 (2014), p. 061801.
- [7] D. Gaiotto, A. Kapustin, N. Seiberg, and B. Willett. “Generalized Global Symmetries”. In: *Journal of High Energy Physics* 2015.2 (2015), p. 172. arXiv: [1412.5148](#).
- [8] H. Georgi. “Effective Field Theory”. In: *Annual Review of Nuclear and Particle Science* 43 (1993), pp. 209–252.
- [9] KamLAND Collaboration, S. Abe, et al. “Precision Measurement of Neutrino Oscillation Parameters with KamLAND”. In: *Physical Review Letters* 100 (2008), p. 221803.
- [10] K. Kraus. “General State Changes in Quantum Theory”. In: *Annals of Physics* 64 (1971), pp. 311–335. DOI: [10.1016/0003-4916\(71\)90108-4](#).
- [11] G. Lindblad. “On the Generators of Quantum Dynamical Semigroups”. In: *Communications in Mathematical Physics* 48 (1976), pp. 119–130.
- [12] A. J. Maley. “Complete Physical Interface of the Neutrino Propagation Role Complex: Carrier-Neutral Invariants, Lawful Equivalence, and Constraint Closure”. PCR-2 publication manuscript. 2026.
- [13] A. J. Maley. “Global Form and Generalized Symmetry of the Standard Model”. 2026.
- [14] A. J. Maley. “Non-Degenerate Construction and the Kernel of Admissibility: An Exhaustion Theorem for Standing, Reference, and Irreversibility”. 2026.
- [15] A. J. Maley. “Role-Occupancy Closure under Admissibility: Why Role-Cardinality Matching Is Only the Cardinality Shadow of Compatible Realization”. 2026.
- [16] A. J. Maley. “Source-First Exhaustion of the Physical Carrier Universe: Independent Candidate Generation for the Neutrino Propagation Role Complex”. PCR-3 publication manuscript. 2026.
- [17] A. J. Maley. “Target-Fixed Physical Role Necessity for the Realized Neutrino Oscillation Interface”. PCR-1 publication manuscript. 2026.
- [18] A. J. Maley. “The AMetric Boundary as a Unique Regime-Invariant Constraint on Admissibility-Bearing Construction”. 2026.
- [19] A. J. Maley. “The Realized Physical Interior: Locality, Metric Role, Fields, Particles, Gauge Structure, Mass, Vacuum, Time, Causality, and the Standard Physics Stack under AASC”. 2026.
- [20] A. J. Maley. “The Standard Model Carrier Exhaustion Closure”. 2026.
- [21] A. J. Maley. “The Standard Model Symmetry Atlas and Exhaustion Closure”. 2026.
- [22] P. Minkowski. “ $\mu \rightarrow e\gamma$ at a Rate of One Out of 10^9 Muon Decays?” In: *Physics Letters B* 67 (1977), pp. 421–428. DOI: [10.1016/0370-2693\(77\)90435-X](#).

- [23] M. A. Nielsen and I. L. Chuang. *Quantum Computation and Quantum Information*. 10th Anniversary. Cambridge University Press, 2010.
- [24] SNO Collaboration, Q. R. Ahmad, et al. “Direct Evidence for Neutrino Flavor Transformation from Neutral-Current Interactions in the Sudbury Neutrino Observatory”. In: *Physical Review Letters* 89 (2002), p. 011301.
- [25] W. F. Stinespring. “Positive Functions on C*-Algebras”. In: *Proceedings of the American Mathematical Society* 6 (1955), pp. 211–216. DOI: [10.1090/S0002-9939-1955-0069403-4](https://doi.org/10.1090/S0002-9939-1955-0069403-4).
- [26] D. Tong. “Line Operators in the Standard Model”. In: *Journal of High Energy Physics* 2017.7 (2017), p. 104. arXiv: [1705.01853](https://arxiv.org/abs/1705.01853).
- [27] S. Weinberg. “Baryon- and Lepton-Nonconserving Processes”. In: *Physical Review Letters* 43 (1979), pp. 1566–1570. DOI: [10.1103/PhysRevLett.43.1566](https://doi.org/10.1103/PhysRevLett.43.1566).
- [28] S. Weinberg. “Phenomenological Lagrangians”. In: *Physica A* 96 (1979), pp. 327–340.
- [29] E. Witten. “An SU(2) Anomaly”. In: *Physics Letters B* 117 (1982), pp. 324–328.